

Integration of CFD Bioreactor Models with Cell Growth Models for the Scale-Up of Cultivated Meat

By

KIRA KIVIAT
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Approved:

Harishankar Manikantan, Co-Chair

David Block, Co-Chair

Jennifer Curtis

Committee in Charge

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Abstract

Cultivated meat, produced by culturing animal cells in bioreactors, has the potential to be a sustainable and ethical method of meat production. However, scaling up these processes to commercially viable levels remains a significant challenge for the burgeoning industry. Achieving this may require culturing animal cells in bioreactors with volumes of 200,000 L or more—an order of magnitude beyond any current animal cell culture processes. Due to the high costs and risks of experimental testing at such scales, computational modeling is an appealing approach for evaluating cell growth and de-risking scale-up.

This thesis presents computational fluid dynamics (CFD) simulations of bioreactors at scales ranging from 200 L up to 200,000 L, conducted using the open source software OpenFOAM. These simulations show that by maintaining constant power-to-volume ratio (P/V) and superficial gas velocity, similar hydrodynamic conditions can be achieved across scales. While these results suggest an acceptable environment for cells grown in suspension, the Kolmogorov length scale may pose a threat to cells grown on microcarriers or in large aggregates.

To quantify the impact of hydrodynamic forces, models for eddy damage as well as bubble bursting damage were integrated into a CFD-based compartment model at each bioreactor scale. Additionally, preliminary work on nutrient-driven growth models for muscle cells is presented. Initial fits were unable to conclusively distinguish between growth limitations due to confluence and fibroblast growth factor 2 (FGF2) concentration. However, further experiments revealed that FGF2 does not exhibit a Monod-like relationship with cell growth. Instead, FGF2 levels must be within a certain range to sustain cell growth, interestingly with 2.5% of the typical initial concentration yielding similar growth of murine C2C12 cells.

The framework developed in this work integrates cell growth and hydrodynamic damage models with CFD simulations, providing insights into the effects of scale-up on cell viability and growth. These findings offer valuable guidance for the design and operation of large-scale bioreactors, helping to understand how cell growth will be impacted as bioreactors are scaled up for cultivated meat production.

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1 Introduction

There is a need for a more sustainable approach to meat production. In terms of environmental impacts, animal agriculture accounts for at least 16.5% of GHG emissions (Twine, 2021) as well as significant air and water pollution. Other public health issues include antibiotic resistance and zoonotic diseases. Since 1940, more than 50% of zoonotic diseases are associated with agriculture (Rohr et al., 2019). There are also issues of animal welfare, with the vast majority of food animals raised in factory farm conditions (Anthis, 2019) and billions of land animals slaughtered each year (Sanders, 2018). Despite this, global meat demand continues to rise.

One solution to this problem is to develop better ways of satisfying this demand with fewer harmful effects. Plant-based meats have grown considerably in the last few years, but so far have not significantly displaced conventional meat consumption. Cultivated meat is an emerging technology with the potential to address the growing demand for meat in a more sustainable and ethical way. By growing real animal cells in bioreactors, it is possible to produce meat without slaughter. Life cycle assessments suggest cultivated meat has the potential to have a carbon footprint much lower than for conventional beef and pork production and comparable to that of conventional chicken production, while having much lower land use and air pollution (Sinke et al., 2023). It has the potential to essentially be a drop-in replacement for conventional meat from a consumer perspective.

Current conventional meat production is on the order of 10^{11} kg/year (Humbird, 2021). Estimates of the number of cells needed to produce one kg of meat range from $\sim 2.9 \times 10^{11}$ cells/kg (Allan, De Bank, and Ellis, 2019) to 10^{12} cells/kg (L. Chen et al., 2022). Hence, to even reach 1% of conventional meat production, cultivated meat would require $\sim 10^{23}$ cells/year. To achieve these high quantities of cells, it will be important for the process to be highly efficient. The cells should ideally be grown at very high density to minimize footprint and building materials.

The cost of producing these cells must also be significantly lowered in order for cultivated meat to be cost-competitive with conventional meat. A recent techno-economic analysis (Negulescu et al., 2023) considered a few scenarios for cultivated meat production, with a target of producing 10^8 kg/year, approximately equivalent to one slaughterhouse in the US. They found that using larger bioreactors led to significantly lower costs, primarily due to requiring fewer facilities. While the cost

of goods sold depends significantly on the cost of cell growth media, for stirred tanks it was only at the >200,000 L scale that the cost reached price parity with conventional beef for any cost of media. This suggests that it may be necessary to produce cells for cultivated meat bioreactors at the 200,000 L scale or even larger.

While microbial culture has been done at that scale, animal cells are more fragile than microbes. Currently, animal cells have not been grown commercially beyond the 25,000 L scale. This is in the biopharmaceutical industry, where the product quantities needed are much lower and the prices of the product are much higher compared to food. As a result, there has not been much pressure to scale up another order of magnitude. The biopharmaceutical industry also does not use species or types of cells (e.g., muscle and fat) which are applicable to cultivated meat.

Experimenting with bioreactors at large scales is extremely costly. Computational fluid dynamics (CFD) simulations are routinely used in industry to analyze bioreactors to predict and optimize bioprocess performance, often by determining whether hydrodynamic conditions are in an acceptable range for the target cell line. In addition, a variety of cell growth models, from Monod models (Xing, Bishop, et al., 2010) to detailed metabolic models (Yongky et al., 2015) and population balance models (Pigou et al., 2017), have been developed. However, these studies have focused on a few cell types that are commonly used industrially, such as Chinese Hamster Ovary (CHO) cells.

More recently, there has been a growing interest in integrating cell growth models with CFD models to incorporate hydrodynamic effects on cell growth. This thesis will begin with background theory and an overview of recent relevant work. Chapter 2 will outline various considerations for cells in bioreactors, followed by an overview of methods used for CFD modeling of bioreactors and recent applications, followed by an overview of compartment modeling and other approaches to integrating cell growth into bioreactors. Chapter 3 will show results of CFD simulations of stirred tank bioreactors ranging from 200 L up to 200,000 L under reasonable operating conditions for animal cell culture, comparing hydrodynamic environments across scales. Chapter 4 will show results of cell growth modeling for cell types relevant to cultivated meat as well as the integration of growth models with CFD bioreactor simulations, including additional models of hydrodynamic damage. Chapter 5 will summarize the results in this thesis as well as opportunities for further work.

2 Background and state of the art

2.1 Microcarriers vs Suspension Culture

Currently, commercial large-scale animal cell culture mainly consists of biopharmaceutical production. This industry commonly uses CHO cells, which have been adapted to grow as individual cells in suspension. In contrast, cultivated meat will require culturing muscle cells which are generally adherent. Adherent cells can be grown in stirred tank or airlift bioreactors by attaching them to small beads called microcarriers (Blüml, 2007). They could potentially also be grown in suspension as cell aggregates, similar to human pluripotent stem cells (Ashok et al., 2016).

Current literature on cultivated meat cell culture generally assumes that the cells will be grown on microcarriers (Cantarero-Rivera et al., 2024; Xueliang Li et al., 2020; Zhang et al., 2021), and the few examples in the literature of culture of cultivated meat cells in spinner flasks use microcarriers. However, it may be possible to adapt the cells to grow in single-cell suspension like CHO cells. This has been done for chicken fibroblasts (Pasitka et al., 2022) and apparently by other companies in the space as well. This is beneficial since removing microcarriers reduces cost and process complexity. However, adapting adherent cells to suspension can be time-consuming and unreliable (Bellani et al., 2020).

Additionally, the mechanisms of cell damage differ between suspension and microcarrier culture (W. Hu, Berdugo, and Jeffrey J. Chalmers, 2011), as will be discussed in Section 2.2. Due to the uncertainty about which approach commercialized cultivated meat will use, this thesis will consider both scenarios.

2.2 Sources of Cell Damage in Bioreactors

2.2.1 Stress from Agitation

A number of reviews (Jeffrey J Chalmers, 2015; Villiger et al., 2018) have pointed out that the idea of animal cells being vulnerable to shear stresses—stress parallel to the cell membrane—in typical operating conditions is something of a "myth". Instead, most observed cell death at high agitation is likely due to bubble damage (Kunas and Papoutsakis, 1990; Villiger et al., 2018) (see Section 2.2.2).

Studies specifically investigating the levels of shear stress that are deadly to CHO cells have found varying results depending on the experimental conditions. By pumping cells through a capillary tube, Vickroy, Lorenz, and W. Kelly (2007) investigated the effect of temporary exposure to different levels of shear stress. They found an increase in cell death after temporary exposure to a max shear of 104 Pa, but no damage at 52 Pa. Cell death generally increased with increasing shear, although they found a surprising decrease in cell death around 225 Pa. A lethal threshold of ~ 32 Pa was found for CHO cells by Neunstoecklin et al. (2015), who used a bioreactor connected to a loop that created a particular level of shear stress in order to study the effect of repeated exposure to high stress levels. They also considered mouse hybridoma (Sp2/0) cells and found a lethal threshold of ~ 25 Pa. Interestingly, they also found improved CHO cell growth at exposure to lower shear (15-21 Pa), though not for the Sp2/0 cells. The specifics of how stress affects cells can vary between cell types, and the thresholds that are damaging likely depend on the duration and frequency of exposure to the stress.

Most previous works have considered CHO cells grown in suspension. The impact of stress may be different for adherent cells as well as for different cell types. For instance, Luo et al. (2011) found that, for mesenchymal stem cells, proliferation was reduced at only 1.5 Pa of laminar shear stress. Cells attached to microcarriers are less able to move and deform with the flow, so shear stresses can be more damaging. Cells attached to microcarriers also can be torn off at high shear stress, which is potentially lethal.

While shear stress can impact cells, it is only one aspect of hydrodynamic stress. A broader measure is the energy dissipation rate (EDR), generally given in units of W/kg or W/m³. The energy dissipation rate captures all of the energy lost through motion of the liquid. This is a more complete representation of the potential for hydrodynamic damage to the cells. CHO cells have been found to withstand acute dissipation rates up to $10^4 - 10^5$ W/kg (Ma, Koelling, and Jeffrey J. Chalmers, 2002). However, as with shear stress, the duration and frequency of exposure are important. Lower levels of dissipation can be damaging to cells if they are exposed over an extended period of time. For CHO cells, a chronic dissipation rate of 290 W/kg has been found to be sufficient to reduce cell growth (Godoy-Silva, Mollet, and Jeffrey J. Chalmers, 2009), and lower levels can have other sub-lethal effects such as reduced growth or changes in morphology or differentiation. Levels as low as

0.4 W/kg have been shown to decrease monoclonal antibody production in CHO cells (Sieck et al., 2013). Unlike in most biopharmaceutical applications where the product is a molecule produced by the cells, in cultivated meat the cells themselves are the product. Thus, only cell growth really matters. However, it will be important to investigate the specifics of how a target cell line responds to different levels of energy dissipation or shear stress, since they could cause physiological or metabolic changes that may impact flavor or nutritional content, as well as growth and differentiation.

2.2.2 Bubble Damage

Bubbles may be the primary source of damage to cells in bioreactors, although there are some different considerations between suspension and microcarrier culture. Cell attachment to rising bubbles is a mechanism that affects both types of culture, as both cells and microcarriers can become attached to bubbles. The bubble's rise can create shear forces high enough to be lethal to the attached cells. In the case of microcarrier culture, this force can tear cells off of the microcarrier, which can be lethal (Bauer et al., 2000).

Another mechanism of bubble damage is bubbles bursting at the surface, which can potentially dissipate enough energy to be lethal to nearby cells. As noted in the previous section, CHO cells are able to withstand dissipation rates up to $10^4 - 10^5$ W/kg. Impellers in stirred tanks do not generally create high enough energy dissipation, which is why Kunas and Papoutsakis (1990) found that high agitation did not damage cells when there was no headspace for bubbles to burst into. Bursting bubbles can create lethal levels of energy dissipation. For example, Walls et al. (2017) numerically simulated bubbles bursting and quantified the liquid volume that reached these levels of energy dissipation during the bursting process.

Cells attached to microcarriers seem to be protected from bursting bubble damage, likely due to the small range of the lethal energy dissipation relative to the size of a microcarrier (Bauer et al., 2000). However, if microcarriers are transported into the foam at the top of the tank, the attached cells can become trapped there and starve.

Bubble-related cell damage can be mitigated through the addition of surfactants like Pluronic F68, which act through multiple mechanisms. By reducing the surface tension of the bubbles, the pressure difference across the bubble interface is reduced according to the Young-Laplace equation, leading to

less energy dissipation when the bubble bursts. In addition, the reduced surface tension makes cells less likely to attach to the bubbles (W. Hu, Berdugo, and Jeffrey J. Chalmers, 2011). Pluronic F68 is FDA approved (Khaliq et al., 2023), so it will likely work for use in cultivated meat production. However, surfactants can reduce mass transfer efficiency by stabilizing the bubble interface, which increases the contact time and therefore saturates the fluid around the bubble, decreasing the mass transfer rate. However, this can be counteracted at higher surfactant concentrations due to increased bubble surface area (Sieblist, Jenzsch, and Pohlscheidt, 2013). While reduced mass transfer can also be mitigated with increased agitation and aeration, those create higher stresses which could reach damaging levels. In addition, Pluronic F68 seems to be less effective at higher cell densities (Jeffrey J Chalmers, 2015). Being able to support a high cell density is important for cultivated meat, which needs to produce extremely large quantities of cells. Hence bubble damage is likely to remain a concern for large scale cultivated meat production.

2.2.3 Turbulent Eddies

Stirred tank bioreactors are generally operated in the turbulent regime. Turbulent eddies can be damaging if they are small enough relative to the cells that a cell cannot simply move with the eddies. Viscosity becomes more significant below a minimum size eddy size, and the turbulent kinetic energy is dissipated as heat. The Kolmogorov scale represents the size of the smallest turbulent eddies, and is given by

$$\eta = \left(\frac{\nu^3}{\varepsilon} \right)^{1/4}, \quad (1)$$

where ν is the kinematic viscosity (m^2/s) and ε is the turbulent energy dissipation rate (W/kg). For single-cell suspension of animal cells which are typically $\sim 15 \mu\text{m}$ this is not much of a problem in standard bioreactor conditions. The smallest turbulent eddies in bioreactors are generally larger than individual cells, so the cells will simply move with the flow. However, for cells grown on microcarriers and therefore unable to move freely, it is the scale of the microcarrier that is relevant. Specifically, Croughan, Hamel, and D. I. C. Wang (1987) found that cells begin experiencing damage when the Kolmogorov scale is below a threshold of $\sim 0.5d - 0.6d$, where d is the microcarrier

diameter. For a typical Cytodex I carrier with diameter $180\ \mu\text{m}$, eddy scales can more readily reach sizes small enough to be damaging. This was found by Cantarero-Rivera et al. (2024), who found a Kolmogorov length as low as $60\ \mu\text{m}$ in a simulated 100 mL bioreactor with stirrer bar. If cultivated meat cells end up being produced on microcarriers or potentially in large cell aggregates, it will be important to manage the energy dissipation rate to keep eddy sizes sufficiently large.

2.3 Mass Transfer in Bioreactors

While bubbles can be damaging to cells, aeration is necessary to provide O_2 to the cells and also remove CO_2 produced by the cells. This requires transport of these gases across the bubble-liquid interface. The flux of gas across an interface is related to the concentration difference between the interface C_L^i and the bulk C_L by a mass transfer coefficient, k_L , as in

$$N_A = k_L(C_L^i - C_L). \quad (2)$$

This can also be written in terms of an overall mass transfer coefficient, k_{OL} , and the liquid concentration that would be in equilibrium with the bulk gas, C_L^* , as in

$$N_A = k_{OL}(C_L^* - C_L). \quad (3)$$

However, k_L is usually approximately equal to k_{OL} , as long as the gas solubility is low, i.e., Henry's constant is large. To convert this into a change in the concentration in the liquid, this is multiplied by the volumetric interfacial area a , i.e., the interface surface area per unit volume, leading to

$$\frac{dC_L}{dt} = k_L a (C_L^* - C_L). \quad (4)$$

This describes the transfer of, e.g., O_2 and CO_2 out of or into the gas bubbles in the bioreactor. Since it is not generally possible to experimentally measure this surface area in a bioreactor, in practice the lumped term $k_L a$ is commonly used for measuring oxygen transfer.

The $k_L a$ must be high enough to balance the oxygen uptake rate of the cells so that the oxygen level in the bioreactor can be maintained at the desired level. Tissot et al. (2012) investigated the

k_La requirements of CHO cells grown in orbitally shaken bioreactors. The cells reached a density of 3.5 million cells/mL, and required a k_La of $7 - 9 \text{ h}^{-1}$ to avoid oxygen limitation, depending on the line of CHO cells used. The value of k_La needed will depend on the density of cells as well as the requirements of a particular cell line.

This type of model can be used for CO_2 removal as well as O_2 transfer (Mostafa and Gu, 2003). However, for large-scale high density cell culture, bubbles tend to become saturated with CO_2 , making the k_La no longer a good model. Xing, Lewis, et al. (2017) have proposed an alternative model for CO_2 stripping that accounts for this by incorporating the bubble residence time and CO_2 saturation.

Various correlations have been developed for k_La , with many of the form

$$k_La = K \left(\frac{P}{V} \right)^\alpha v_s^\beta, \quad (5)$$

where v_s is the superficial gas velocity (m/s), P is power (W), V is the liquid volume (m^3) and K, α, β are parameters fitted to experimental data. However, the k_L and a can be calculated separately in CFD simulations. The mass transfer coefficient k_L is commonly calculated according to Higbie's penetration theory as (Hernández-Calderón et al., 2017)

$$k_L = 2\sqrt{\frac{D}{\pi t_c}}, \quad (6)$$

where D is the molecular diffusivity (m^2/s) and t_c is the contact time (s). It is also common (Bach, 2018; Xueliang Li et al., 2020) to take the Kolmogorov time scale $t_c = \left(\frac{\varepsilon}{\nu}\right)^{1/4}$ as the contact time, to get

$$k_L = 2\sqrt{\frac{D}{\pi}} \left(\frac{\varepsilon}{\nu}\right)^{1/4}. \quad (7)$$

The volumetric surface area a can be calculated given the gas volume fraction α_g and the bubble diameter d_b as

$$a = \frac{6\alpha_g}{d_b}. \quad (8)$$

In a stirred tank, the bubble diameter will vary throughout the tank. In models, d_b will either be assumed to be constant or will have a distribution throughout the tank according to a population balance model. The mass transfer coefficient k_La provides a constraint on bioreactor design and choice of operating conditions, since it must be high enough to balance out the cells' oxygen uptake rate. By using CFD, it is possible to calculate this value under different conditions, enabling optimization of processes.

2.4 Types of Bioreactors

There are a number of types of bioreactors in use, many of which could be considered for cultivated meat production. See Cantarero Rivera and J. Chen (2022) for a discussion of bioreactor types in this context. The most widely used type of bioreactor in the biopharmaceutical industry is the stirred tank reactor (Schirmer et al., 2021). As a result, it is well characterized, with well-established processes. This bioreactor type consists of a motor-powered rotating shaft with attached impellers. There are a variety of impeller types in use. A common type is the Rushton turbine, consisting of a horizontal disk surrounded by 6 vertical blades. Rushton impellers are in the class of radial flow impellers, creating flow in a plane perpendicular to the shaft. In contrast, pitched-blade impellers have angled blades which create more axial mixing. These can provide more circulation between the top and bottom of the tank, and are considered a safer option for animal cell culture where shear is more of a concern.

An alternative to stirred tanks are airlift bioreactors. These consist of a tank with a divider creating two sections, one called the riser and one called the downcomer. These sections can be split between the center and the outside or between sides of the tank. Gas is sparged in under the riser, causing flow upwards through that section and back down through the downcomer. Airlift reactors are more energy efficient than stirred tank reactors (Chisti and Moo-Young, 2003). Recent work has suggested that airlift reactors could also be a good candidate for large scale cell culture, including in the context of cultivated meat (Xueliang Li et al., 2020). They have been found to have improved mass transfer (Nadal-Rey, Dale D. McClure, Kavanagh, Cassells, et al., 2022) compared to stirred tanks, and create a more uniform culture environment (Xueliang Li et al., 2020). Negulescu et al. (2023) also found the best cost-effectiveness in a scenario using airlift reactors. However, unlike

stirred tanks, it is not possible to control the mixing and the aeration separately. Since they rely entirely on bubbles for mixing, bubble-related damage may be more of a concern.

Another proposed bioreactor type for cultivated meat is the hollow fiber bioreactor (Allan, De Bank, and Ellis, 2019). A hollow fiber bioreactor consists of a bundle of hollow fibers made of a material that is permeable to oxygen and other nutrients. The bundle is typically housed in a cylindrical cartridge, and a bioreactor can consist of multiple such cartridges. The inside of the fiber (or membrane) is referred to as the lumen, and the space outside of the fibers is the extra-capillary space (ECS). This structure allows for a very large surface area for mass exchange in a compact structure, while also providing a protected environment for the cells (Eghbali et al., 2016). This is particularly beneficial for mammalian cells, which are relatively difficult to culture due to their sensitivity to shear stress and oxygen diffusion. (Allan, De Bank, and Ellis, 2019) argue that hollow fiber bioreactors are a good option for cultivated meat due to the potential for very high density culture. This is important since manufacturing bioreactors is a major initial cost, making entry into the industry at scale difficult (Humbird, 2020), and there are also environmental costs associated with manufacturing bioreactors. Thus, being able to use fewer materials by growing cells at high density makes hollow fiber bioreactors a promising approach for cultivated meat. However, they do not scale as cost-effectively as stirred tanks and have never been scaled up to the levels needed for cultivated meat production. Harvesting the cells by detaching them from these membranes at scale would be highly inefficient, as would cleaning and sanitizing the large membrane surface area between uses.

Due to the ubiquity of stirred tank reactors, they are still likely to be used in commercial cultivated meat production. As a result, this thesis will focus on that type of bioreactor. However, extending this work to other types of bioreactors will be a valuable next step.

2.5 CFD Methods for Bioreactors

Given that the main uncertainties in scaleup of cultivated meat lie in the mass transfer and shear forces that may be present at commercial scale, CFD modeling of these scales is an important tool to de-risk the design of these large-scale fermentors. Here, we discuss the considerations for this type of CFD approach.

2.5.1 Multiphase Models

Aeration is important for animal cells in bioreactors, so it is generally necessary to model the gas as well as the liquid in the bioreactor. There are two common approaches to modeling a liquid / gas system: Euler-Euler, in which both phases are treated as overlapping continua, and Euler-Lagrange, in which the gas bubbles are modeled as individual particles in the continuous fluid phase. Euler-Euler is generally what is used for bioreactor modeling, while Euler-Lagrange is limited to small scale reactors with low gas holdup, since the large scale and number of bubbles makes tracking individual bubbles computationally impractical (Besagni, Inzoli, and Ziegenhein, 2018).

In Euler-Euler, a phase fraction α_k is used to represent the fraction of volume made up of each phase k , subject to

$$\sum \alpha_k = 1. \quad (9)$$

Each phase also satisfies a continuity equation

$$\frac{\partial}{\partial t}(\alpha_k \rho_k) + \nabla \cdot (\alpha_k \rho_k \vec{u}_k) = 0 \quad (10)$$

and a momentum equation

$$\frac{\partial}{\partial t}(\alpha_k \rho_k \vec{u}_k) + \nabla \cdot (\alpha_k \rho_k \vec{u}_k \vec{u}_k) = -\alpha_k \nabla p + \nabla \cdot \underline{\underline{\tau_k}} + \alpha_k \rho_k \vec{g} + \vec{M}_k. \quad (11)$$

Here $\vec{M}_k$ represents the mass transfer between the phases, consisting of drag, lift, virtual mass, turbulent dispersion, and wall lubrication forces (Besagni, Inzoli, and Ziegenhein, 2018). Among these forces, drag is generally considered the most important, so the others might be neglected (Cappello et al., 2021).

Euler-Euler models are typically used for modeling bioreactors, including stirred tanks (e.g., Nadal-Rey, Dale D. McClure, Kavanagh, Cassells, et al. (2022), Nauha, Kálal, et al. (2018), and Villiger et al. (2018)) as well as airlift reactors and bubble columns (e.g., Xueliang Li et al. (2020), Nadal-Rey, Dale D. McClure, Kavanagh, Cassells, et al. (2022), and Rahimi et al. (2018)) due to their reasonable accuracy and relatively low computational cost. For the same reason, this is the approach

that will be used in this work.

2.5.2 Bubble Models

Multiphase models require the specification of a bubble size or sizes. In airlift or bubble column reactors this is straightforward, particularly since cell culture media generally does not support much coalescence. As a result, the bubble sizes follow a fairly uniform distribution (D. D. McClure et al., 2013). According to Klaas van’t Riet and Tramper (2014), the equilibrium bubble diameter in coalescing media is 6 mm. As a result, a single bubble size is typically used for bubble columns, e.g., Jolius Gimbut (2009) and Dale D. McClure et al. (2014).

For stirred tank reactors, the bubble size distribution is less uniform since the impeller breaks up bubbles, creating smaller bubbles. As a result, some studies have used population balance models for bubble sizes in stirred tanks, which include models of bubble breakup and coalescence. Nonetheless, other studies have achieved reasonable predictions while assuming a constant bubble size (Cappello et al., 2021; Jamshidian, Scully, and Van den Akker, 2023; Scargiali et al., 2007).

Among papers modeling bubbles in stirred tanks using a constant bubble diameter, there is a fairly wide range of diameters used, from 0.5 mm to 7 mm. Some (like Scargiali et al. (2007)) have found that varying bubble diameter does not make much difference for gas holdup since the gas distribution primarily depends on the bubble terminal velocity, which does not vary significantly over the range of bubble sizes that might occur in a bioreactor.

Values used for modeling stirred tanks in the literature using a single bubble size (or pressure-adjusted) are shown in Table 1. While bubble size may not strongly influence the gas holdup, it will affect the k_La by changing the bubble surface area. However, since the gas distribution is not heavily impacted, it is possible to change the bubble size after the simulation is complete in order to compute the k_La , or even to apply a population balance model to the bubbles after the simulation.

2.5.3 Impeller Rotation

Stirred tank reactors are very common in industrial use, and they perform their mixing via rotating impellers. While it is possible to use a moving mesh to simulate the actual rotation of the impeller, a popular alternative is the multiple reference frame (MRF) model. In MRF, a rotation region is

Table 1: Bubble diameters used in the literature among studies that used a single bubble size while modeling a stirred tank reactor. All of these are specific to bioreactors except X. Hu et al. (2021) and J. Gimbut, Rielly, and Nagy (2009), which are more general.

source	bubble diameter	justification	scale
Jamshidian, Scully, and Van den Akker (2023)	1.8, 2.8 mm	Compared experimental gas holdup after the fact and found 1.8 worked well	2 L
Panunzi et al. (2022)	4.4 mm	Based on correlation from an old paper, using orifice Re (so ignoring breakup from impeller)	4100 L
Nadal-Rey, Dale D. McClure, Kavanagh, Cassells, et al. (2022)	5 mm (at outlet, adjusted for pressure)	D. D. McClure et al. (2013), measurements for bubble column	90,000 L
Zhang et al. (2021)	2 mm	based on experiments	250 mL
X. Hu et al. (2021)	0.5, 1.5, 2.5 mm	found best agreement with literature data on gas holdup using 0.5 mm bubbles	7 L
Hajian, Zieringer, and Takors (2021)	7 mm	says based on other papers, but those used population balance models	54,000 L
Cappello et al. (2021)	4.1 mm	experimental measurements	22,000 L
Farzan and Ierapetritou (2018)	2.89 mm	Cite Alves et al. (2002)	3 L
J. Gimbut, Rielly, and Nagy (2009)	3.5 mm	compared results to BSD data from Laakkonen et al. (2007)	200 L
Scargiali et al. (2007)	2, 3, 4 mm	compared simulation results and found little difference	70 L

defined around the impeller and the velocity is calculated at each point in that region relative to the rotating frame, $\vec{u}_r$, in terms of the velocity relative to the stationary frame, $\vec{u}$, the rotation vector, $\vec{\Omega}$, and the vector from the rotation center to the point, $\vec{r}$:

$$\vec{u}_r = \vec{u} - (\vec{\Omega} \times \vec{r}). \quad (12)$$

Plugging this into the (incompressible, steady) momentum equation leads to an additional momentum source term:

$$\nabla \cdot (\vec{u}\vec{u}_r) = -\frac{\nabla p}{\rho} + \nu \nabla^2 \vec{u} - (\vec{\Omega} \times \vec{u}). \quad (13)$$

This approach is much less computationally intensive than the moving mesh, and generally provides a reasonable steady state approximation of the flow as long as the flow is roughly uniform around the edge of the rotating zone (Peng et al., 2019). It is less useful if there are significant asymmetries in the geometry, such as a sparger tube on one side (Xianhua Li et al., 2018). Nonetheless, the MRF approach is often a worthwhile trade-off for bioreactor modeling due to the greater computational efficiency (Scargiali et al., 2007), and is what will be used here.

2.5.4 Turbulence Models

Flow in agitated bioreactors is generally at high Reynolds number, so the flow is turbulent. Capturing the full details of turbulent fluctuations using direct Navier-Stokes requires an extremely fine spatial and time resolution, making it infeasible for bioreactor spatial and temporal scales. Instead, turbulence models can be used to capture major effects of turbulence without resolving such fine detail.

Typically, Reynolds Averaged Navier-Stokes (RANS) is used for modeling bioreactors due to its relative computational efficiency. In this model, the velocity is written as the sum of the average velocity $\bar{U}_i$ and fluctuating components u'_i . Plugging this into the Navier-Stokes equations, they can be written in terms of average velocity but with an additional fluctuation term, the "Reynolds stresses". According to the Boussinesq Hypothesis, these fluctuation terms can be related to the

mean velocity in terms of an eddy viscosity ν_t as

$$-\overline{u'_i u'_j} = \nu_t \left(\frac{\partial \overline{U}_i}{\partial x_j} + \frac{\partial \overline{U}_j}{\partial x_i} \right) - \frac{2}{3} k \delta_{ij}, \quad (14)$$

where $k = \frac{1}{2} \overline{u'_i u'_i}$ is the turbulence kinetic energy and δ_{ij} is the Kronecker delta. Then it is just necessary to model this eddy viscosity, ν_t . A number of closure models for this exist. The most commonly used model for bioreactors is the k - ε model, in which the turbulent kinetic energy, k , and the turbulent energy dissipation rate, ε , are modeled by solving transport equations for each. From these, the eddy viscosity can be calculated as $\nu_t = C_\mu \frac{k^2}{\varepsilon}$, where $C_\mu \approx 0.09$ is an empirically-determined constant. An alternative, which is better at capturing near-wall effects, is the k - ω model, which models the specific dissipation rate $\omega = \frac{\varepsilon}{k}$ instead of ε directly. However, this model has its own shortcomings. An alternative closure model which combines the two is the shear stress transport (SST) k - ω model, which essentially functions like k - ω near the wall, and k - ε in the free stream. This model has been found to be slightly better than k - ε for bubbly flow (Besagni, Inzoli, and Ziegenhein, 2018). However, it may work less well with a multiple reference frame (MRF) approach to impeller rotation (Section 2.5.3), as found by (Lane, 2017). The k - ε model is still widely used in the literature for modeling stirred tank reactors.

An alternative approach to turbulence modeling is large eddy simulation (LES), which directly simulates the flow down to a certain scale, and then uses a sub-grid scale model below that. This approach is better at capturing details of fast fluctuations (Besagni, Inzoli, and Ziegenhein, 2018). However, RANS is usually used instead due to the significantly lower computational cost and the fact that RANS has been found to be a reasonably good approximation for bioreactor modeling (W. J. Kelly, 2008), so RANS will be used in this work.

2.5.5 Software

A large number of papers using CFD for bioreactors use commercial software. Many use Ansys software, usually Ansys Fluent (e.g., Farzan and Ierapetritou (2018), Hajian, Zieringer, and Takors (2021), Jossen, D. Eibl, and R. Eibl (2021), Xueliang Li et al. (2020), Morchain, Gabelle, and Cockx (2014), and Zhang et al. (2021)), and sometimes Ansys CFX (e.g., Nadal-Rey, Dale D. McClure,

Kavanagh, Cassells, et al. (2022) and Scargiali et al. (2007)), which are finite volume solvers. Some work has used the finite element solver COMSOL (e.g., Cantarero-Rivera et al. (2024)), although it is less specialized for fluids. More recently, there has been an increase in Lattice-Boltzman solvers for CFD, in part because of their ability to use GPUs effectively. The software M-Star is designed around this, and has been used recently for bioreactor simulations (Haringa, 2023; Šrom et al., 2024).

While commercial software packages are fully-featured and supported, some research uses the open source software OpenFOAM (e.g., X. Hu et al. (2021) and Rahimi et al. (2018)). Due to its open nature, researchers can build solvers on top of its infrastructure for specific applications. The research consortium Helmholtz-Zentrum Dresden-Rossendorf have published an extension to OpenFOAM with a focus on multiphase flow (Schlegel et al., 2024), although they are geared primarily towards bubble columns.

OpenFOAM was used for the simulations in this work. The open source nature of the software makes it possible to fully share any work done with the tool in a way that it can be run by anyone without the need for an expensive license.

2.6 Related Applications of CFD to Bioreactors

Recent CFD studies on the hydrodynamics of large scale bioreactors have primarily focused on microbial cell culture, since prior to cultivated meat there has not been strong pressure to scale up animal cell culture beyond 20,000 L. For instance, Nadal-Rey, Dale D. McClure, Kavanagh, Cassells, et al. (2022) evaluated both stirred tank and airlift reactors for microbial culture at scales up to 90,000 L. They used an Euler-Euler multiphase model and the $k-\varepsilon$ turbulence model in Ansys CFX. Using a constant bubble size modified by pressure, they found superior mixing performance in the bubble column configuration. Another example is Rahimi et al. (2018), who simulated airlift and bubble column reactors at scales nearing 500,000 L. They also employed an Euler-Euler multiphase model and the $k-\varepsilon$ turbulence model, but used OpenFOAM as their software.

While most large-scale simulations target microbial systems, some studies have examined animal cell culture. Villiger et al. (2018) simulated bioreactors ranging from 3 L to 15,000 L using Ansys Fluent. Their work validated key metrics such as $k_L a$ and maximum shear stress against experimental

measurements. However, their simulations did not approach the 200,000 L scale that may be required for cultivated meat production.

Research explicitly tailored to cultivated meat bioreactors remains limited. A recent review by Cantarero Rivera and J. Chen (2022) outlines various types of bioreactors and modeling approaches, as well as potential approaches to integrating CFD cell growth models. One notable study by Xueliang Li et al. (2020) simulated a 300,000 L airlift bioreactor for cultivated meat, assuming microcarrier-based cell culture. Using Ansys Fluent software with an Euler-Euler multiphase model and $k-\varepsilon$ turbulence model, they concluded that the airlift reactor could support cell densities of 2×10^8 cells/mL.

Experimental studies have also begun exploring bioprocess development for cultivated meat. Two recent studies used spinner flasks to culture bovine myoblasts (Verbruggen et al., 2018) or bovine adipose-derived stem cells (Hanga et al., 2021). Building on this, Zhang et al. (2021) used Ansys Fluent to simulate spinner flasks under conditions similar to those studies and scaled their CFD simulations to a 20,000 L bioreactor. They found that sufficient agitation to keep microcarriers suspended resulted in Kolmogorov length scales below the damage threshold for cells on microcarriers in significant portions of the tank. Cantarero-Rivera et al. (2024) used COMSOL and also simulated a small bioreactor intended for microcarrier culture for cultivated meat. They too found significant regions with Kolmogorov length scale below the damaging threshold, although they only simulated a 100 mL bioreactor and neglected aeration.

Most work on CFD simulations of bioreactors has either been large-scale but focused on microbial culture, or it has focused on animal cell culture or even cultivated meat, but only at relatively small scales. There remains a clear need for work that evaluates bioreactors under conditions suitable for animal cells at the scales required for cultivated meat production (Negulescu et al., 2023).

2.7 Compartment Modeling

Evaluating the hydrodynamic conditions in a bioreactor can provide insight into bioreactor design, but to really understand how cells will be affected by the conditions in a bioreactor, it is useful to integrate models of cell growth with models of the hydrodynamic conditions. However, animal cell culture processes can take days or longer, and simulating that time span with CFD is

computationally impractical.

Compartment modeling can overcome the limitation of only being able to simulate a snapshot or short time period of a bioprocess. In CFD-based compartment modeling, a fluid simulation is performed, and then the tank is divided into regions of similar flow. Each compartment is taken to have uniform values of gas holdup, hydrodynamic stresses, and other features, and the flux between boundaries of compartments is calculated to determine the movement of cells between adjacent compartments. Using these compartments, it is possible to run a much faster simulation of a bioprocess by starting with some concentration of cells and nutrients in each compartment, and updating over time by simulating growth and death models within each compartment as well as flux between compartments. This increased efficiency makes it more feasible to use complex metabolic models as well.

Compartments can be defined manually based on knowledge of the flow field structure. For Rushton stirred tank reactors, this is often done by defining nested cylinders, with separate regions around the impellers (e.g., Farzan and Ierapetritou (2018) and Nauha, Kálal, et al. (2018)). Alternatively, the compartments can be calculated automatically by clustering adjacent cells with similar flow values (Tajsoleiman et al., 2019), as done, e.g., by Nadal-Rey, Dale D. McClure, Kavanagh, Cassells, et al. (2021). This method works by clustering computational cells in a 2D plane and rotating the selection around the tank. Besides CFD-based compartment approaches, there have been efforts to develop compartmental models based on data from flow-following sensors (Bisgaard et al., 2021).

The compartment modeling process can be extended to account for varying bioreactor fill levels over the course of a batch process. For example, Farzan and Ierapetritou (2018) simulated bioreactors at different fill levels, and switched the compartment model from one base simulation to another over the course of the simulated process. Nadal-Rey, Dale D. McClure, Kavanagh, Cassells, et al. (2021) did something similar, although the process of mapping from one compartment model to the next was made more complicated by the use of automatic compartments. They also generated intermediate fill volumes in between those actually simulated with CFD by cutting off the top parts of the uppermost compartments from the next size up that was actually simulated.

Compartment modeling enables rapid simulation of a cell culture process, although it does come

at the cost of significant simplifications in the hydrodynamics. However, the trade-off is likely worthwhile. Nadal-Rey, Dale D. McClure, Kavanagh, Cassells, et al. (2021) compared compartment simulations with full CFD simulations and found comparable results for mixing time as well as similar results for the integrated growth model of *S. cerevisiae*. These similar results were achieved in a massively shorter time frame (2 s on 1 core versus 30 days on 24 cores). Compartment modeling is a promising approach to integrating cell growth models with hydrodynamic simulations, but so far it has not been applied to cultivated meat.

2.8 Alternative Approaches to Integrating Cell Growth in Bioreactors

Besides compartment modeling, there have been other approaches in the literature to integrating cell growth models with bioreactor environments. One of the simplest approaches is exemplified by O’Brien et al. (2021). In this model, the bioreactor environment is treated as a single, well-mixed compartment, interacting with a metabolism model and a cell growth model. Bulk concentrations of nutrients and metabolites, including gas exchange due to sparging, are tracked in the bioreactor environment and influence cell growth. However, this method assumes a perfectly uniform environment in the bioreactor and neglects hydrodynamic effects such as shear stress or nutrient gradients as well as pressure gradients due to hydrostatic head which will be significant at large scales.

More sophisticated approaches include Euler-Lagrange methods, which solve the fluid phase(s) with an Eulerian model while tracking cells or particles as Lagrangian elements (Hajian, Zieringer, and Takors, 2021; Haringa, Tang, G. Wang, et al., 2018; Lapin, Müller, and Reuss, 2004). This method provides detailed insights into the environment experienced by individual cells over time, capturing variations in stress and nutrient availability at a granular level. However, the computational cost is prohibitively high for resolving long time scales, making it unsuitable for processes spanning days, as required for cell culture.

Cantarero-Rivera et al. (2024) applied Euler-Lagrange methods to track microcarriers instead of individual cells, focusing on their distribution within the bioreactor. They found that microcarriers tended to accumulate in a high stress region above the impeller. Unlike simulating a full cell growth process, the microcarrier distribution can be determined over a much shorter time scale. They also took a novel approach by combining CFD modeling with agent-based models to evaluate

stresses experienced by cells on microcarriers. They developed an agent-based model of cells on a microcarrier, accounting for changes to cell proliferation and death due to stresses, and combined this with the fluid flow to investigate the effect of agitation on cell growth on a microcarrier. While this approach can provide a detailed understanding of hydrodynamic stresses, at present they only simulated a single microcarrier and assumed a fixed cell doubling time, omitting nutrient dynamics and their impact on cell growth. This limits its applicability for simulating a full cell growth process.

Another alternative is the population balance model used by Morchain, Gabelle, and Cockx (2014). They used an Euler-Euler approach to solve the gas and liquid phases, while tracking cell populations as passive scalars divided into different growth rate classes. The transition of cells between classes depends on the difference between the class specific growth rate and the equilibrium growth rate with the local nutrient concentrations. This approach captures the lag in growth rate adjustment due to environmental changes, which is particularly relevant in large bioreactors where nutrient gradients can be significant (Nadal-Rey, Dale D. McClure, Kavanagh, Cornelissen, et al., 2021).

While more detailed approaches like population balance and Euler-Lagrange modeling provide a granular view of bioreactor dynamics, they come with significant computational costs. In contrast, the compartment modeling approach described in Section 2.7 offers a practical balance between simplicity and accuracy. By dividing the reactor into compartments, each with distinct nutrient concentrations and growth conditions, this method accounts for spatial gradients without the computational intensity of fully-resolving fluid-cell interactions. Although it assumes instantaneous growth rate changes on movement of cells between compartments, it allows for rapid simulation of growth models with varying parameters.

Moreover, compartment modeling is flexible and can be extended by integrating elements from more complex methods, such as combining it with population balance models or even Lagrangian tracking (Haringa, Tang, and Noorman, 2022). This adaptability, coupled with its computational efficiency, makes compartment modeling an appealing choice for studying large-scale bioreactors while leaving room for future enhancements.

3 CFD modeling of bioreactors¹

3.1 Abstract

Cultivated meat has the potential to mitigate harmful effects from conventional meat production. However, it will likely require animal cells to be produced in bioreactors at an order of magnitude larger scale than has been done so far. To help de-risk this scale up, simulations of plausible bioreactor configurations were simulated at a series of scales ranging from 200 L to 200,000 L using computational fluid dynamics. Several different drag models were compared, and the one that predicted the lowest mass transfer and highest shear was used in order to be conservative about the predicted cell environment. The distributions of shear stress, oxygen mass transfer coefficient, and Kolmogorov length scale were compared across the bioreactor scales, showing only minor changes with increasing scale. The case of a Rushton and pitched impeller was compared to a case with two Rushton impellers, and the latter was found to have higher mass transfer and only slightly higher shear for a given power input. This study provides a step towards predicting animal cell culture performance at the scales needed for cultivated meat production.

3.2 Introduction

Currently industrial animal cell culture is primarily limited to low volume high value biopharmaceutical production, and has only been performed at 20 m³ scale commercially (Humbird, 2021). However, cultivated meat may need to be grown in significantly larger reactors (~ 200 m³) to be viable, based on the scales used to produce products through microbial fermentation at similar volumes or price points (Xueliang Li et al., 2020) and technoeconomic analyses (Negulescu et al., 2023). It is costly to run experiments at large scale, so computational modeling is a useful tool to aid in assessing the feasibility of growing muscle cells at that scale, as well as in optimizing reactor design, media composition and feed strategy, and cell line development.

Most existing investigations into large scale culture and fermentor design focus on microbial fermentation. Hydrodynamic stresses are more relevant to animal cells than to microbes, which is why animal cell culture typically uses lower agitation and aeration rates and different impeller types designed to limit shear. Chinese hamster ovary (CHO) cells have been found to be sensitive to

¹An adapted version of the contents of this chapter is being written up for publication in a peer-reviewed journal.

energy dissipation rates that may occur in large scale reactors (Sieck et al., 2013). Stresses can come from the agitation itself, as well as from gas sparging, which is necessary to provide the cells with sufficient oxygen.

Stirred tank bioreactors are generally operated in the turbulent regime, and cells can be damaged by turbulent eddies. The Kolmogorov length scale is a measure of the smallest-scale turbulent eddies. This is most relevant to cells on microcarriers, rather than in suspension. If the eddies are smaller than the microcarrier size, cells can be damaged and pulled off of the microcarriers. Freely suspended cells, on the other hand, will just move with the eddies (Croughan, Hamel, and D. I. C. Wang, 1987). Cells that are relevant for cultivated meat are generally adherent. While there have been efforts to adapt these kinds of cells to suspension culture (Pasitka et al., 2022) to avoid the additional cost and complexity of using microcarriers, in practice these cells tend to form clumps when grown in suspension, so they may still behave similarly to cells on microcarriers.

Bubbles are likely the primary source of cell damage, as animal cells have been found to handle high agitation rates as long as there is no sparging or headspace into which bubbles burst (Kunas and Papoutsakis, 1990). Both microcarriers and individual cells can become attached to rising bubbles, which can expose the cells to damaging shear. The effects of bubble damage can be reduced by the addition of surfactants like Pluronic F68, as opposed to through fermentor design, although this tends to reduce oxygen transfer (Sieblist, Jenzsch, and Pohlscheidt, 2013).

Animal cell growth depends on sufficient dissolved oxygen levels. CHO cells have been found to be sensitive to low dissolved oxygen levels (Ozturk and Palsson, 1991) and high CO₂ (Brunner et al., 2017). In very large reactors (on the order of 10 m tall), there will be a significant pressure gradient between the top and bottom of the reactor (nearly 1 atm), which will affect solubility of these gasses (Villiger et al., 2018). The increased oxygen and CO₂ solubility at the bottom of the fermentor should help increase oxygen transfer, but also CO₂ which has been shown to inhibit growth of some animal cell lines (Gray et al., 1996). The oxygen transfer coefficient, k_La , is a measure of how effectively oxygen can be transported from bubbles into the culture medium. Experimental measurements of k_La generally evaluate volume averages, but CFD enables calculating a spatial distribution of k_La based on the gas distribution and bubble size.

Most previous modeling work has focused on biopharmaceutical-relevant cell lines. Recent work such as Nadal-Rey, Dale D. McClure, Kavanagh, Cassells, et al. (2022), Nauha, Kálal, et al. (2018), and Rahimi et al. (2018) all simulated large-scale ($\sim 100 - 500 \text{ m}^3$) airlift or stirred tank bioreactors, but these were done with an eye towards microbial culture. This is reflected in the operating conditions, including the use of only Rushton impellers and a high agitation rate in the stirred tank cases. Very limited work has examined conditions relevant to cultivated meat production. Villiger et al. (2018) did look at conditions in bioreactors under animal cell culture conditions. However, they only examined up to the 15,000 L scale. Farzan and Ierapetritou (2018) attempted to integrate animal cell growth models into a CFD simulation of a bioreactor, but only at $\sim 3 \text{ L}$ scale. For cultivated meat production, it will be important to simulate the conditions predicted for commercial production ($>200 \text{ m}^3$), which is an order of magnitude larger than any existing cell culture production system, in order to de-risk the purchase and installation of this type of system. Work that is directly relevant to cultivated meat is even more limited. Zhang et al. (2021) simulated a stirred flask that has been successfully used for growing cultivated meat-relevant cells, and then simulated a 20 m^3 bioreactor to compare the environment to the flask. Xueliang Li et al. (2020) evaluated mass transfer and mixing performance in a 300 m^3 airlift reactor in the context of cultivated meat. Cantarero-Rivera et al. (2024) focused on the dynamics of cell attachment to microcarriers and the stresses experienced by cells on microcarriers. These studies all assumed the cells would be grown on microcarriers, and none simulated the most common type of commercial bioreactor system—the stirred tank—at the scale envisioned for cultivated meat.

The goal of this work is to compare multiple scales of stirred tank bioreactors at a range of plausible operating conditions, focusing on key features relevant to animal cell culture such as k_La and shear stress. Similar bioreactor geometries were examined at four different scales under typical animal cell culture conditions, including the current industrial max of $\sim 20 \text{ m}^3$ up to a target of 200 m^3 . The distribution of energy dissipation, k_La , and Kolmogorov length scale were evaluated for each case to get an understanding of how the scale may affect culture of cultivated meat cells. A comparison between impeller types was also performed at the 200 m^3 scale to examine the difference between the multi-Rushton configuration typical of microbial culture with the Rushton and pitched blade configuration typical of animal cell culture.

3.3 Methods

3.3.1 Geometry and Meshing

A similar geometry was used for all the bioreactor scales investigated here, ranging from 200 L to 200,000 L working volume. A representative image of the geometry is in Figure 1. It consists of a Rushton impeller at the bottom and a pitched blade impeller (“Elephant Ear” style) above with blades at a 45° angle, a ring sparger, and four baffles. All the geometries were programmatically generated in the open-source software Salome using a Python script with a configuration file specifying the dimensions.

The dimensions were chosen to be consistent with a typical bioreactor used for animal cell culture. Animal cell culture generally uses a lower tank height to diameter ratio of 2:1, rather than 3:1 more typical of microbial culture. Animal cell culture also tends to use larger diameter impellers, approximately half the tank diameter. A typical spacing of 1.5 impeller diameters between adjacent impellers, and 1 impeller diameter between the lowest impeller and the tank bottom, was used, with the sparger at half the height of the lowest impeller. The baffle width and wall gap match that used in Bombač et al. (1997).

Table 2: Dimensions used for each geometry. Everything is related to the tank diameter, which is set as needed at each scale to obtain the desired volume.

H_F / T	B / T	g / T	D_s / D_i	D_t / D_s	H_s / D_i	H_i / D_i	S / D_i
1.6	1/12	1/60	0.85	0.08	0.5	1	1.5
D_i / T	H / D_i	D_d / D_i	f / T	a / D_i	b / D_i	L / D_i	H_c / D_i
0.4	0.2	0.85	0.05	0.15	0.2	0.68	0.8

For simplicity of meshing, the sparger was modeled as an annular cylinder with the entire top surface treated as a gas inlet. The exact shape of the sparger likely does not matter much, since virtually all of the air will be pulled through the bottom impeller and distributed based on the agitation.

For the meshing, a base hexahedral cylinder mesh was generated with the OpenFOAM utility `blockMesh`. Then another OpenFOAM utility, `snappyHexMesh`, was used to refine the mesh to fit the geometry STLs generated by Salome.

Since the majority of hydrodynamic stresses in a stirred tank occur around the impeller blades, the mesh was refined more around each impeller. Layers were added along all the walls to ensure

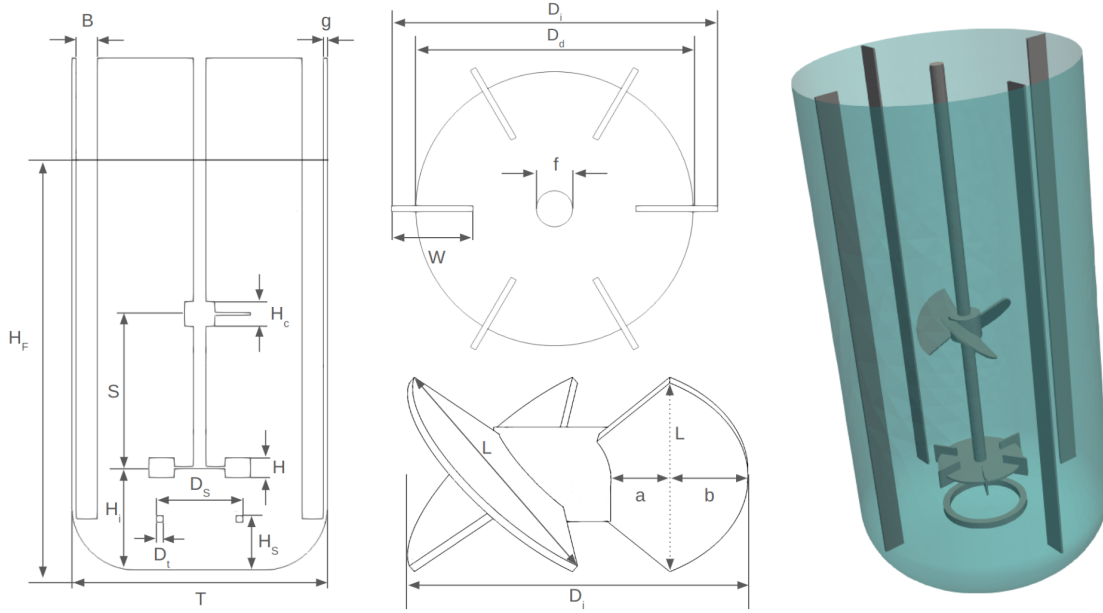


Figure 1: Geometry used at all scales. H_F : fill height, T : tank diameter, B : baffle width, g : baffle gap, D_s : sparger outer diameter, D_t : sparger tube thickness, H_s : sparger top height, H_i : height to center of bottom impeller, S : spacing between impeller centers, D_i : impeller diameter, D_d : impeller disk diameter, W : Rushton blade width, H : Rushton blade height, f : shaft diameter, L : pitched blade corner-to-corner length, a : distance from inner blade edge to line L , b : distance from line L to outer blade edge

that the y^+ values were generally in the log-law region of $30 < y^+ < 500$ (Greenshields and Weller, 2022), with mean values ~ 300 , and the vast majority of the area ($>80\%$) below 300.

The simulation results should converge to a consistent solution as the mesh becomes finer. If the mesh is too coarse, refining it will change the results. To ensure the solutions here are not dependent on the refinement level, simulations were performed for each bioreactor scale at a base mesh size and one with $\sim 50\%$ more cells, and the results compared to ensure they did not change significantly with the mesh size (Figure 5). The mesh refinement was defined by the cell size in the base hexahedral mesh, and was increased with scale slightly less than proportional to the increase in geometry size, to ensure low y^+ values at the largest scales. The largest scale mesh (200 m^3) required additional wall layers to obtain a sufficiently low y^+ .

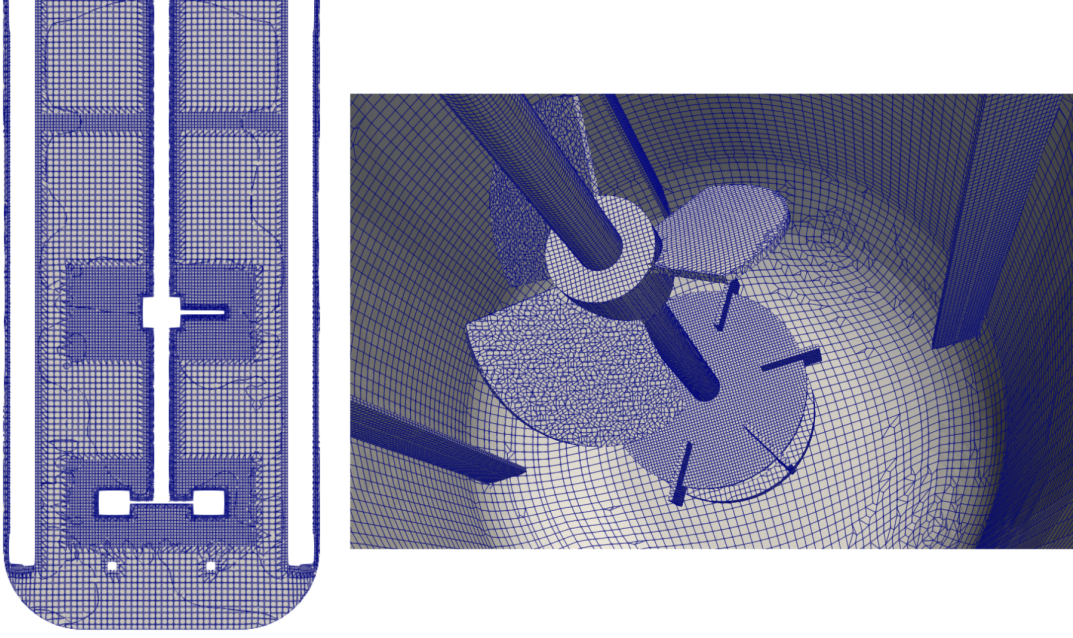


Figure 2: Example mesh with 720,877 cells

3.3.2 Multiphase Model

The gas and liquid phases were modeled with a compressible Euler-Euler multiphase solver (the OpenFOAM `multiphaseEulerFoam` solver). This type of solver treats the phases as overlapping continua with shared pressure, solving a set of mass and momentum conservation equations for each phase. This approach does not track individual bubbles or interfaces, instead solving for a volume fraction of gas at each location. This makes it computationally less expensive than simulating individual bubbles, which is particularly important for very large scale reactors which may have millions of bubbles.

The mass and momentum conservation equations for each phase i are

$$\frac{\partial \alpha_i \rho_i}{\partial t} + \nabla \cdot (\alpha_i \rho_i \vec{U}_i) = 0, \quad (15)$$

$$\frac{\partial \alpha_i \rho_i \vec{U}_i}{\partial t} + \nabla \cdot (\alpha_i \rho_i \vec{U}_i \vec{U}_i) = -\alpha_i \nabla p + \nabla \cdot (\alpha_i \vec{\tau}_i) + \alpha_i \rho_i \vec{g} + \vec{M}_{ij}, \quad (16)$$

where p is the pressure, g is gravity, $\alpha_i, \rho_i, \vec{U}_i, \vec{\tau}_i$ are, respectively, the volume fraction, density,

velocity, and the Newtonian viscous stress tensor of phase i , and $\vec{M}_{ij}$ is the momentum transfer between phases i and j (e.g., l for liquid phase and g for gas phase).

The momentum transfer may consist of drag, lift, wall force, turbulent dispersion, and virtual mass, with drag typically being the most important (Cappello et al., 2021). In particular, lift, wall force, virtual mass, and turbulent dispersion can generally be neglected for stirred reactors (Scargiali et al., 2007), although X. Hu et al. (2021) argue that turbulent dispersion may actually be important. Following their work, in this study, drag, virtual mass, and turbulent dispersion were used. Hence the momentum transfer term can be expressed as

$$\vec{M}_{ij} = \vec{F}_D + \vec{F}_{VM} + \vec{F}_{TD}, \quad (17)$$

where $\vec{F}_D$ is the drag force, $\vec{F}_{VM}$ is the virtual mass force, and $\vec{F}_{TD}$ is the turbulent dispersion force.

3.3.2.1 Drag The most significant interphase momentum transfer term is the drag force imparted on the gas bubbles by the liquid. The hydrodynamic drag on the gas phase can be written as

$$\vec{F}_{D,g} = \frac{3}{4} \rho_l C_D \frac{\alpha_g}{d_g} (\vec{U}_l - \vec{U}_g) |\vec{U}_l - \vec{U}_g|, \quad (18)$$

where d_g is the particle diameter and C_D is the drag coefficient. The force on the liquid phase has the opposite sign, $\vec{F}_{D,l} = -\vec{F}_{D,g}$.

There are several models commonly used for the drag coefficient. A common one is the Schiller Naumann drag model (e.g., Jolius Gimbut (2009) and Zhang et al. (2021)), implemented in OpenFOAM as `SchillerNaumann`, which is given by

$$C_D = \begin{cases} \frac{24}{Re} (1 + 0.15 Re^{0.687}) & Re < 1000 \\ 0.44 & Re > 1000 \end{cases}, \quad (19)$$

using the bubble Reynolds number $Re = \frac{\rho_l d_g |\vec{U}_g - \vec{U}_l|}{\mu_l}$.

Another common model is Tomiyama (Tomiyama et al., 1998) (e.g., X. Hu et al. (2021), Xueliang Li et al. (2020), and Nauha, Kálal, et al. (2018)), which, as implemented in the `TomiyamaCorrelated` drag model in OpenFOAM, is given by

$$C_D = \max \left\{ A \cdot \min \left[\frac{1}{Re} (1 + 0.15 Re^{0.687}), \frac{3}{Re} \right], \frac{8}{3} \frac{E_0}{E_0 + 4} \right\}, \quad (20)$$

where Eo is the Eötvös number, relating gravity to surface tension for a bubble: $Eo = \frac{\Delta \rho g L^2}{\gamma}$, with $\Delta \rho$ the density difference between the phases, γ the surface tension and L the characteristic length (e.g., the bubble radius of curvature).

The coefficient A can have different values depending on the contamination of the liquid, e.g., with surfactants. $A = 16$ corresponds to a pure system and $A = 24$ corresponds to a slightly contaminated system. This accounts for the fact that surfactants immobilize the interface, which increases drag (Manikantan and Squires, 2020).

A third model commonly used is Ishii-Zuber (Ishii and Zuber, 1979) (e.g., Nadal-Rey, Dale D. McClure, Kavanagh, Cassells, et al. (2021) and Shi and Rzehak (2018)), implemented in OpenFOAM as `IshiiZuber`, which is more complex and can account for non-spherical bubbles. All three models were evaluated in this work.

3.3.2.2 Virtual Mass Virtual mass occurs when a particle accelerates relative to the surrounding fluid, as the particle must push the fluid out of the way. It is proportional to the mass of fluid displaced by the particle multiplied by the relative acceleration. This force was modeled with a constant coefficient C_{VM} , so the force on the particle is given by

$$\vec{F}_{VM,g} = C_{VM} \alpha_g \rho_l \left(\frac{D(\vec{U}_l - \vec{U}_g)}{Dt} \right), \quad (21)$$

where $\frac{D}{Dt}$ is the material derivative, and $C_{VM} = 0.5$ is the coefficient for a sphere, i.e., the virtual mass force is proportional to half the mass displaced by the sphere (per volume, as with all the forces presented here) (Crowe et al., 2011). The force on the liquid is the negative of this term.

3.3.2.3 Turbulent Dispersion Turbulent dispersion occurs due to interactions between the bubbles and turbulent eddies. It was modeled using the Lopez de Bertodano model (Lopez de Bertodano, 1992):

$$\vec{F}_{TD} = C_{TD}\alpha_g\rho_l k_l \nabla\alpha_g, \quad (22)$$

where k_l is the turbulent kinetic energy of the liquid and the constant coefficient $C_{TD} = 1.0$.

3.3.2.4 Bubble Diameter Bubbles were modeled using a constant size adjusted for pressure. While air bubbles in water show significant coalescence, leading to a wide distribution of bubble sizes, the size distribution has been found to be much narrower in cell culture media (D. D. McClure et al., 2013). Hence, it is common to use a single bubble size when modeling bioreactors, e.g., Huang et al. (2018) and Nadal-Rey, Dale D. McClure, Kavanagh, Cassells, et al. (2022).

Assuming the sparger is below the impeller, virtually all of the bubbles will go through the impeller. Bubbles disperse at the vortices behind impeller blades, producing very small bubbles. If the cell culture media composition promotes coalescence due to addition of antifoam or other surfactants in the media, the bubbles will coalesce to reach an equilibrium size based on the turbulent forces in the bulk, typically 3-6 mm for stirred vessels (Klaas van't Riet and Tramper, 2014). For all scales, the bubble diameter was set to be 4 mm at the outlet (at atmospheric pressure). Some studies, such as Villiger et al. (2018), have used population balance models, which account for coalescence and break up of bubbles, leading to a distribution of bubble sizes. However, the increased complexity and computation time may be unnecessary. In preliminary simulations (data not shown), as well as in other studies such as Scargiali et al. (2007), variation in the bubble diameter does not make much difference in the gas distribution. The bubble size will affect the calculated k_La by changing the interfacial area per unit volume. However, since the gas distribution does not vary significantly, the bubble size used for calculating k_La could be adjusted after the simulation is complete.

3.3.3 Turbulence Model

A RANS turbulence model with the $k-\varepsilon$ two-equation model was chosen for this study. This is a commonly used model for bioreactor modeling, providing a reasonable trade-off between accuracy and efficiency, although the RANS approach is known to underestimate stresses. Specifically, the

Table 3: Physical properties for the liquid and gas phases

liquid phase	molecular weight	18 g/mol
	ρ	993 kg/m ³
	c_p	4183 J/(kg K)
	μ	0.0007 Pa s
	Pr	4.73
gas phase	molecular weight	28.9 g/mol
	c_p	1006 J/(kg K)
	μ	1.89×10^{-5} Pa s
	Pr	0.7

mixture k - ε model (Behzadi, Issa, and Rusche, 2004) was used, which solves a single set of equations for both phases in terms of mixture quantities. This model can work for any phase fraction, unlike models that just assume the continuous phase always dominates the turbulence. The equations look like

$$\frac{\partial(\rho_m k_m)}{\partial t} + \nabla \cdot (\rho_m \vec{U}_m k_m) = \nabla \cdot \frac{\mu_m^t}{\sigma_m} \nabla k_m + P_k^m - \rho_m \varepsilon_m + S_k^m, \quad (23)$$

$$\frac{\partial(\rho_m \varepsilon_m)}{\partial t} + \nabla \cdot (\rho_m \vec{U}_m \varepsilon_m) = \nabla \cdot \frac{\mu_m^t}{\sigma_m} \nabla \varepsilon_m + \frac{\varepsilon_m}{k_m} (C_{\varepsilon 1} P_k^m - C_{\varepsilon 2} \rho_m \varepsilon_m) + C_{\varepsilon 3} \frac{\varepsilon_m}{k_m} S_k^m. \quad (24)$$

See Behzadi, Issa, and Rusche (2004) for more details.

3.3.4 Fluid Properties

Although cell culture media is not the same as water, it is similar enough in density and viscosity to typically be treated as having the physical properties of water for CFD simulations of bioreactors. In this study, the liquid phase properties were based on those of water at 37°C (Dinger and Zamfirescu, 2015), and the gas phase properties were based on air at 37°C. The surface tension was set to 70 mN / m, although in reality it may be lower in a bioreactor due to surfactants in the media.

The relevant physical properties are summarized in the following tables, where c_p is the specific heat and $\text{Pr} = \frac{c_p \mu}{k_{tc}}$ is the Prandtl number, representing the ratio of momentum diffusivity to thermal diffusivity, where k_{tc} is the thermal conductivity (W/m K). The gas phase used the perfect gas model to determine the density.

3.3.5 Boundary Conditions

While it is common to use a “degassing” boundary condition at the liquid surface, (Nadal-Rey, Dale D. McClure, Kavanagh, Cassells, et al., 2022) found that it underpredicted absolute pressure. Hence the simulations in this paper instead included a gas headspace at the top of the tank, with the entire top surface of the geometry treated as an outlet for both phases. Walls were treated as no-slip for the liquid phase and slip for the gas phase, except for the impeller shaft outside of the MRF zone, which was modeled as having rotational velocity for the liquid phase. Wall functions were used for k and ε on all walls. The upper surface of the sparger was treated as an inlet for the gas phase and a slip wall for the liquid phase.

3.3.5.1 Velocity Inlet Values While the entire sparger surface was used as an inlet, the volume fraction of air at the inlet was calculated as the fraction of the surface area that is open (based on number and size of orifices). The same area fraction of 4.4% was used at each scale, taken to match that in Bombač et al. (1997). The velocity of the inlet gas was then calculated to achieve the target superficial gas velocity at the inlet.

3.3.5.2 Turbulence Inlet Values Turbulence intensity is the ratio of velocity fluctuations to velocity, given by

$$I = \frac{\vec{U}'_{RMS}}{|\vec{U}|}, \quad (25)$$

where

$$\vec{U}'_{RMS} = \left(\frac{\overline{\vec{U}'^2}}{3} \right)^{1/2} = \sqrt{\frac{\overline{u'u'} + \overline{v'v'} + \overline{w'w'}}{3}} = \sqrt{\frac{2k}{3}} \quad (26)$$

since the turbulent kinetic energy is $k = \frac{1}{2}(\overline{u'u'} + \overline{v'v'} + \overline{w'w'})$ for $\vec{U} = (u, v, w)$.

Hence $I = \frac{\sqrt{2k/3}}{|\vec{U}|}$, and so

$$k = \frac{3}{2}(|\vec{U}|I)^2. \quad (27)$$

This value of k is used as the inlet as well as initial value, using the impeller tip speed as the reference

velocity $|\vec{U}|$. The choice of I depends on the strength of the turbulence, commonly between 1% and 5%. In this work, I was taken to be 1%, corresponding to low turbulence.

The inlet dissipation was calculated as

$$\varepsilon_{\text{in}} = \frac{C_\mu \rho k^2}{\mu \mu_r}, \quad (28)$$

where $C_\mu = 0.09$ and μ_r is the ratio of eddy viscosity to fluid viscosity. For low turbulence, μ_r was set to be 1.

3.3.6 Multiple Reference Frame Model

The multiple reference frame (MRF) approach was used to model the rotation of the impeller. This approach is much less computationally intensive than the moving mesh, and generally provides a reasonable steady state approximation of the flow as long as the flow is roughly uniform around the edge of the rotating zone (Peng et al., 2019). It has been found to be a good trade-off typically for stirred tank bioreactors (Amer, Feng, and Ramsey, 2019), as long as there are not significant asymmetries in the geometry (Xianhua Li et al., 2018).

The size of the MRF zone should be chosen to be as large as possible without reaching the point where the influence of the wall or baffles is reaching the MRF zone, as this leads to the most realistic results (Patil et al., 2021). MRF may be insufficient if there are internal components too close to the impeller, as in Nadal-Rey, Dale D. McClure, Kavanagh, Cassells, et al. (2022), but in this work the MRF approach appears to be sufficient as the flow field is approximately uniform around the outside of the MRF zone. For this work, the MRF radius was set to $0.3T$, where T is the tank diameter.

3.3.7 Numerics and Convergence

The simulations were run until the volume-averaged values had largely stabilized, then averaged over an additional 30 seconds. In most cases, by 60 seconds, the volume average values of liquid velocity, k , ε , and gas holdup had stabilized, and the gas flux at the outlet corresponded to the sparger gas inlet flow (accounting for volume change due to the pressure difference). However, averaging was

started later at 70 seconds for the largest case, as the values took longer to stabilize.

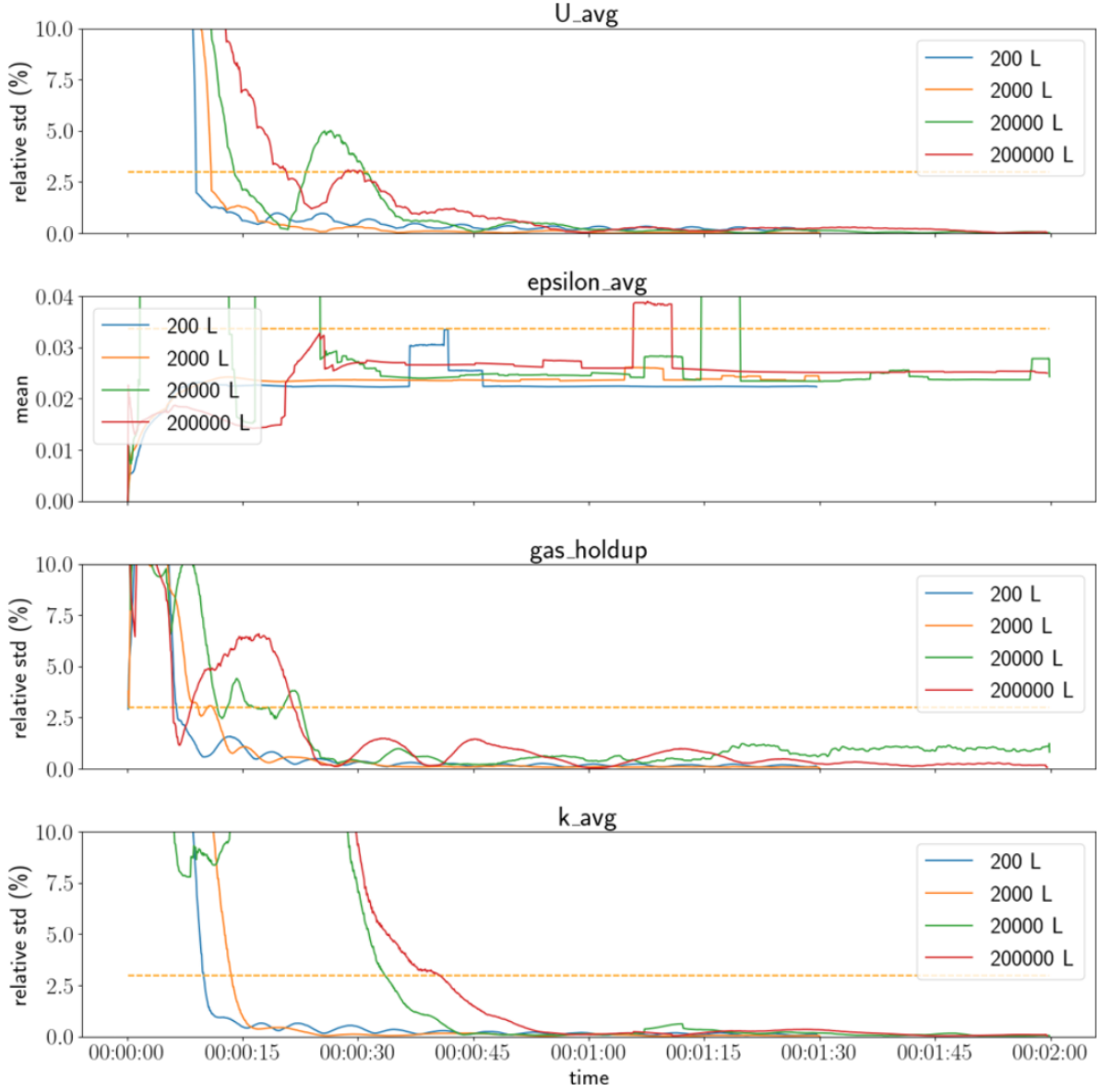


Figure 3: Plots showing convergence of volume-averaged field values over simulation time. The values are a rolling relative standard deviation over a 5 second window in all cases except epsilon, which shows a rolling mean instead due to very brief large spikes in maximum epsilon values over the course of the simulation.

3.3.8 Operating Conditions

The aeration rate affects the gas distribution in the tank, which can determine whether cells in the reactor have enough oxygen to thrive. It is common to scale aeration rate either by keeping vvm (void volumes per minute = working volumes per minute) constant or by keeping superficial gas

velocity constant. Aeration rates are generally kept low for animal cell culture, generally between 0.005-0.1 vvm since their oxygen uptake rate is an order of magnitude smaller than for microbial systems, and it reduces sparging-associated cell damage (Al-Rubeai, 2015).

Nauha, Visuri, et al. (2015) recommend scaling aeration through constant superficial gas velocity rather than vvm, as the hydrodynamics can vary significantly with constant vvm at different scales. This study used a constant superficial gas velocity of 6.7×10^{-4} m/s at the sparger inlet height, corresponding to approximately 0.01 vvm at the 20,000 L scale.

Input power per volume is another important factor affecting the hydrodynamic conditions in the reactor as well as the mixing quality. The value of P/V is commonly kept constant while scaling to maintain similar conditions.

The gassed power is different from the ungassed power, and the ratio of these is related to the aeration number:

$$N_A = \frac{F}{ND^3}, \quad (29)$$

where F is the volumetric flow rate, N the agitation rate, and D the impeller diameter.

In all cases, the aeration number was below 0.01, so the ratio of gassed to ungassed power should be quite close to 1 (Doran, 2013). Hence the agitation required was set based on the target ungassed P/V ratio.

According to Xueliang Li et al. (2020), typical animal cell culture uses 5-80 W/m³. The agitation rate at all scales was set to keep the P/V constant at ~ 25 W/m³, calculated using

$$N = \left(\frac{P}{N_P \rho D_i^5} \right)^{1/3}, \quad (30)$$

where D_i is the impeller diameter, ρ is the liquid density, P is the target power, and N_P is the power number. This corresponds to impeller tip speeds ranging from 0.8 m/s at the smallest scale to 1.7 m/s at the largest scale. Due to the low aeration level, the ungassed power and the gassed power are approximately equal. The power number was taken to be 5 for the Rushton impeller

based on Geankoplis (2006) (Figure 4), since in all cases the Reynolds number was sufficiently large (from $\sim 75,000$ at the 0.2 m^3 scale to $\sim 1,600,000$ at 200 m^3 scale). A power number of 1.7 was used for the pitched blade impeller, based on Zhu et al. (2009). The impeller power numbers were added to get the total power number (Geankoplis, 2006) under the assumption that the impellers are sufficiently far enough apart to be independent at $1.5D_i$ spacing.

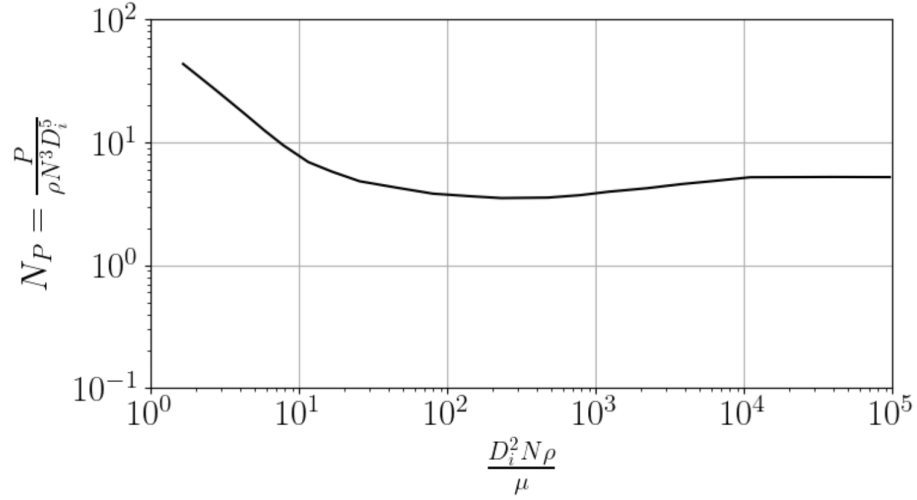


Figure 4: Relationship between power number and Reynolds number. Depicted data correspond to a single Rushton impeller and four baffles. Adapted from Geankoplis (2006).

Table 4: Agitation rates at each scale corresponding to $\sim 25 \text{ W} / \text{m}^3$ power input.

	200 L	2,000 L	20,000 L	200,000 L
agitation rate	69.7 RPM	41.8 RPM	25 RPM	15 RPM
tip speed	0.79 m/s	1.02 m/s	1.32 m/s	1.70 m/s

To verify the choice of agitation rate, the power per volume was estimated from the CFD simulations and was in each case close to $25 \text{ W}/\text{m}^3$. Power was calculated from the simulation by integrating the sum of the turbulent and viscous energy loss for each phase:

$$P = \int [\alpha_l(\rho_l \varepsilon_l + \mu_l S_l^2) + \alpha_g(\rho_g \varepsilon_g + \mu_g S_g^2)] dV \quad (31)$$

where S_i is the strain rate magnitude for phase i , calculated as $\sqrt{2D_i : D_i}$ where D_i is the strain rate tensor.

3.3.9 Measures of Cell Stress

3.3.9.1 Energy Dissipation Energy dissipation rate is a broad measure of stress experienced by cells. The average energy dissipation rate should be related to the P/V as

$$\bar{\varepsilon} = \frac{P}{\rho V} \quad (32)$$

which aligns well with the calculated volume averages of $\varepsilon \approx 0.021 - 0.027$. However, the average value is not sufficient since transient exposure to regions of higher stress can have a detrimental effect on cell growth. It is also important how high the stresses get and how large the volume is above a damaging threshold.

3.3.9.2 Shear Shear is another measure of the stress experienced by cells. According to Soos et al. (2010), the shear stress due to turbulent fluctuations experienced by cells at sub-Kolmogorov length scale can be calculated as

$$\tau_{vs} = \frac{5}{2} \mu \sqrt{\frac{\varepsilon}{6\nu}} \approx \mu \sqrt{\frac{\varepsilon}{\nu}}. \quad (33)$$

3.3.9.3 Kolmogorov Length Scale The Kolmogorov length scale, η , is a relevant consideration since cells may be damaged by eddies below a size threshold. In particular, for cells on microcarriers, this threshold is $\eta < 0.6d_p$ where d_p is the diameter of the microcarrier (from Zhang et al. (2021), citing Nienow et al. (2016)). The Kolmogorov scale is

$$\eta = \left(\frac{\nu^3}{\varepsilon} \right)^{1/4}, \quad (34)$$

where ε is the turbulent energy dissipation rate. Regions with η below the critical threshold have potential for cell damage.

3.4 Results

3.4.1 Evaluating Modeling Methodology and Parameters

3.4.1.1 Mesh Comparison To ensure that model predictions were independent of the mesh refinement, the results were compared to those from a mesh with $\sim 50\%$ more cells. The different

mesh sizes showed similar results, based on volume average values and axial-averaged field values taken in a plane halfway between the impellers (Figure 5). At all scales, even with significantly smaller mesh, it can be seen that model predictions are not impacted. Thus, subsequent simulations were performed using the coarser mesh to reduce computational time.

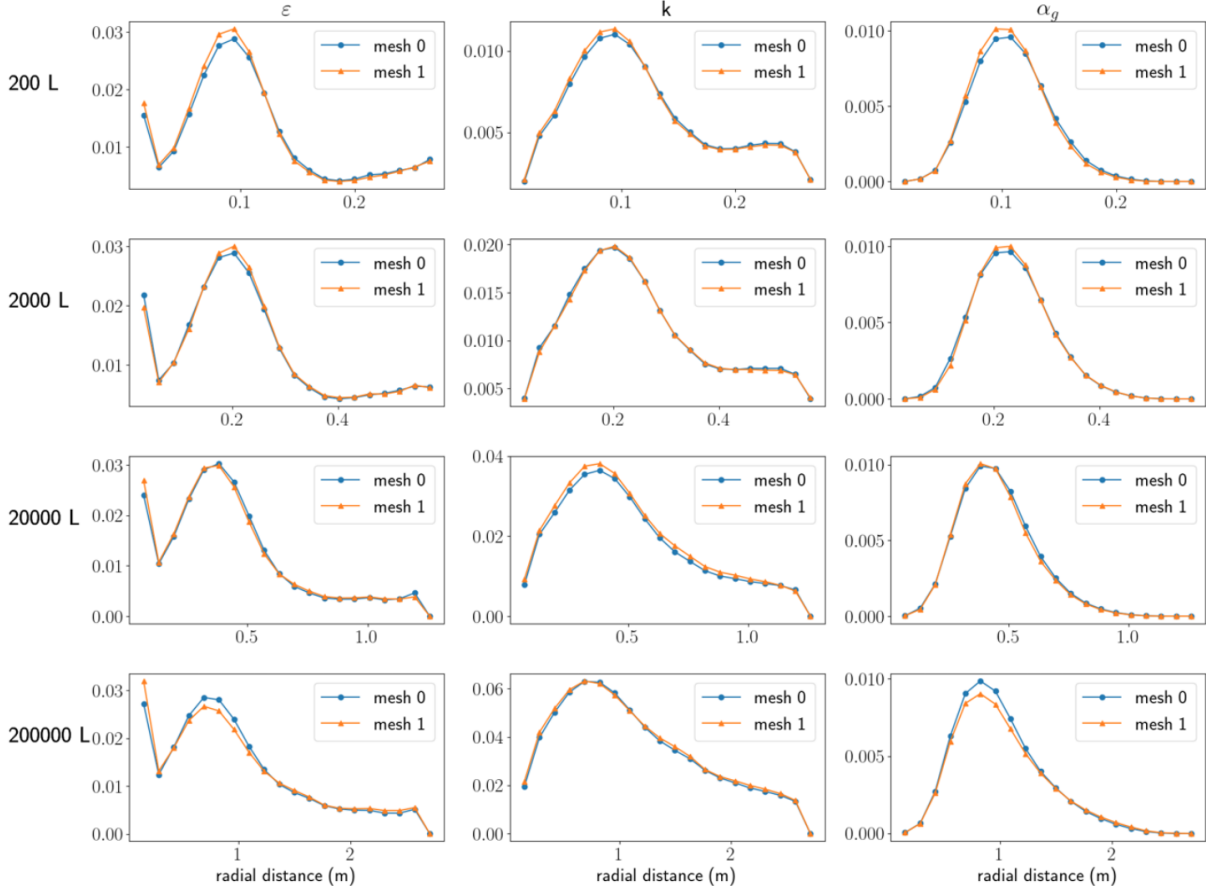


Figure 5: Plots of axial-averaged field values vs radial distance from tank center at height halfway between the impellers, comparing results at baseline size (mesh 0) and refinement (mesh 1). First column is mean turbulent energy dissipation ε (W/m^3), second column is kinetic turbulent energy k (m^2/s^2), third column is gas volume fraction.

3.4.1.2 Drag Model Comparison Preliminary tests were performed at the 200 L scale comparing different drag models (for gas bubbles moving through the liquid phase) to choose one for the subsequent simulations. The models compared were Schiller-Naumann, Tomiyama (corresponding to slight contamination), and Ishii-Zuber, described in the Methods section. The results are shown in Figure 6. The gas distribution and flow field look qualitatively fairly similar across all three models, although with lower gas holdup for Schiller-Naumann. Since Schiller-Naumann predicted

the lowest k_La and the highest shear under the given conditions, that model was used for all further simulations in order to be conservative about k_La and shear predictions.

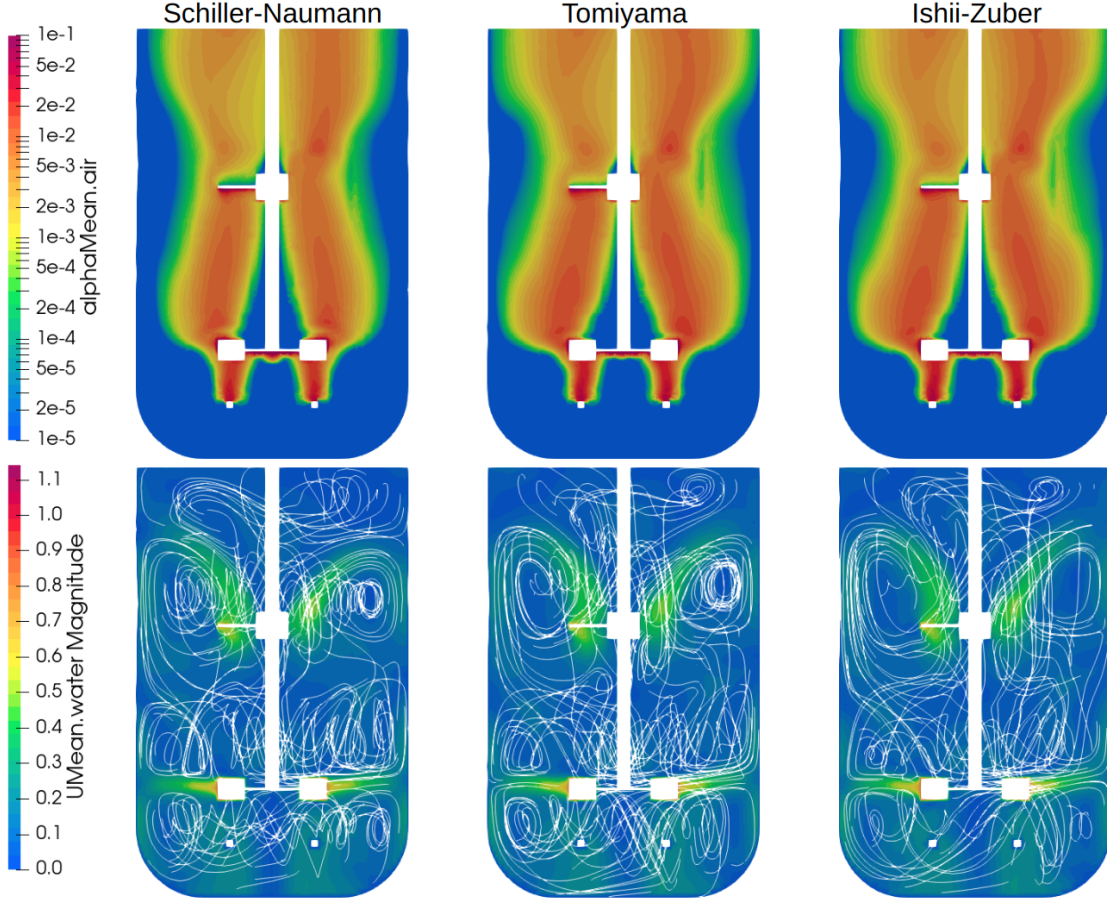


Figure 6: Comparison of time-averaged gas volume fraction and velocity streamlines for three different drag models at 200 L scale.

Table 5: Comparison of time-averaged volume-averaged quantities for different drag models in 200 L geometry.

	kLa	shear	gas holdup	P/V
Schiller-Naumann	14.45	0.1147	0.2478%	41.70
Tomiyama	15.97	0.1110	0.2891%	40.43
Ishii-Zuber	16.67	0.1099	0.3072%	39.65

3.4.2 Model validation

The same geometry generation process was used to create a geometry similar to the 5,000 L bioreactor simulated by Villiger et al. (2018). This was run with the same agitation and aeration rates to compare against their results, which were also validated against experimental data. The volume-

averaged k_La value calculated here for the 5,000 L case was $\sim 4.2 \text{ h}^{-1}$. For comparison, the value calculated from the similar CFD simulation in Villiger et al. (2018) was $\sim 5.0 \text{ h}^{-1}$, and the corresponding experimental data was $\sim 3.6 \text{ h}^{-1}$.

For the same case, the volume distribution of shear stress was calculated and compared to the results in Villiger et al. (2018) (Figure 7). The average shear stress was similar but higher in this work (average of 0.11 Pa for this work, compared to 0.1 Pa reported in Villiger) and the entire distribution skewed to higher shear stresses in our modeling. That is not surprising, since we used a pitched blade impeller while Villiger et al. (2018) used a marine impeller, which generally leads to lower shear. The fact that the results for shear and k_La are qualitatively similar for these similar geometries and operating conditions supports the validity of the simulations presented here and allows us to predict oxygen transfer and shear stresses at significantly larger scales that have not yet been attempted experimentally.

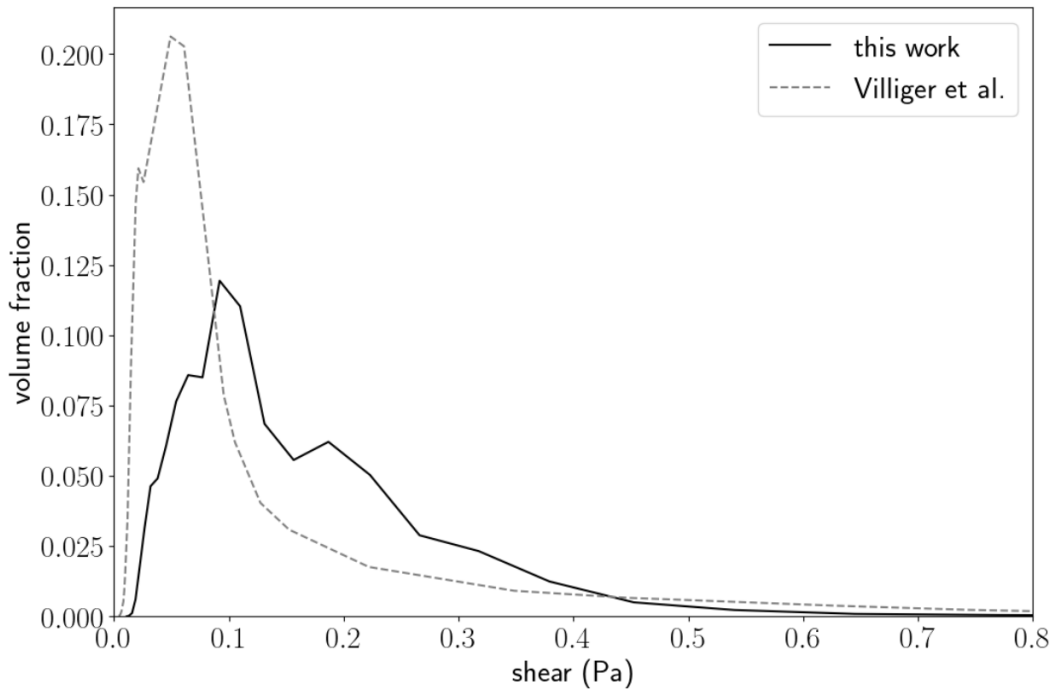


Figure 7: Volume distribution histogram of shear stress in a 5 m³ bioreactor simulation compared with the distribution for a similar case in Villiger et al. (2018)

3.4.3 Impact of scale

Simulations were performed at a range of scales from 200 L to 200,000 L to determine whether there are significant differences across these scales while keeping P/V and superficial gas velocity constant.

3.4.3.1 Mass Transfer Reactor size was scaled keeping P/V and superficial gas velocity constant, which should lead to similar hydrodynamics across the scales. In all cases, the gas holdup was quite low at $\sim 0.2\%$, and Figure 8 shows that the gas was concentrated towards the center of the tank, with essentially no aeration below the sparger. This is reflected in the $k_L a$ as well, although in the log scale it can be seen that there is somewhat more aeration towards the bottom of the tank at the largest volume. The $k_L a$ was calculated by separately estimating k_L according to Higbie’s penetration theory, using the Kolmogorov time scale as the contact time, according to

$$k_L = 2\sqrt{\frac{D}{\pi}} \left(\frac{\varepsilon}{\nu}\right)^{1/4}, \quad (35)$$

where D is the gas diffusivity (taken to be $2 \times 10^{-9} \text{ m}^2/\text{s}$ as for gas in water), and calculating the volumetric surface area as

$$a = \frac{6\alpha_g}{d}. \quad (36)$$

The volume average $k_L a$ was similar at all scales, ranging from $7.6 - 8.2 \text{ h}^{-1}$, without a strong trend across scales. To ensure oxygen is distributed, the mixing time should be well below the characteristic time $1/k_L a$. The mixing time can be estimated using the following equation (van der Lans and K. Riet, 2011)

$$t_m \approx \frac{10}{N} \frac{\left(\frac{T}{D}\right)^2}{N_P^{1/3}} \left(\frac{H_L}{T}\right)^2, \quad (37)$$

where H_L is the liquid height, T is the tank diameter, D is the impeller diameter, N is the agitation rate, and N_P is the power number. This gives values ranging from 20 s at the 200 L scale to 95 s at the 200,000 L scale. In all cases, this is well below the characteristic time ranging from $1/k_L a \approx 415\text{s} - 460\text{s}$.

For CHO cells in orbitally shaken bioreactors, Tissot et al. (2012) found the cells required a $k_L a$

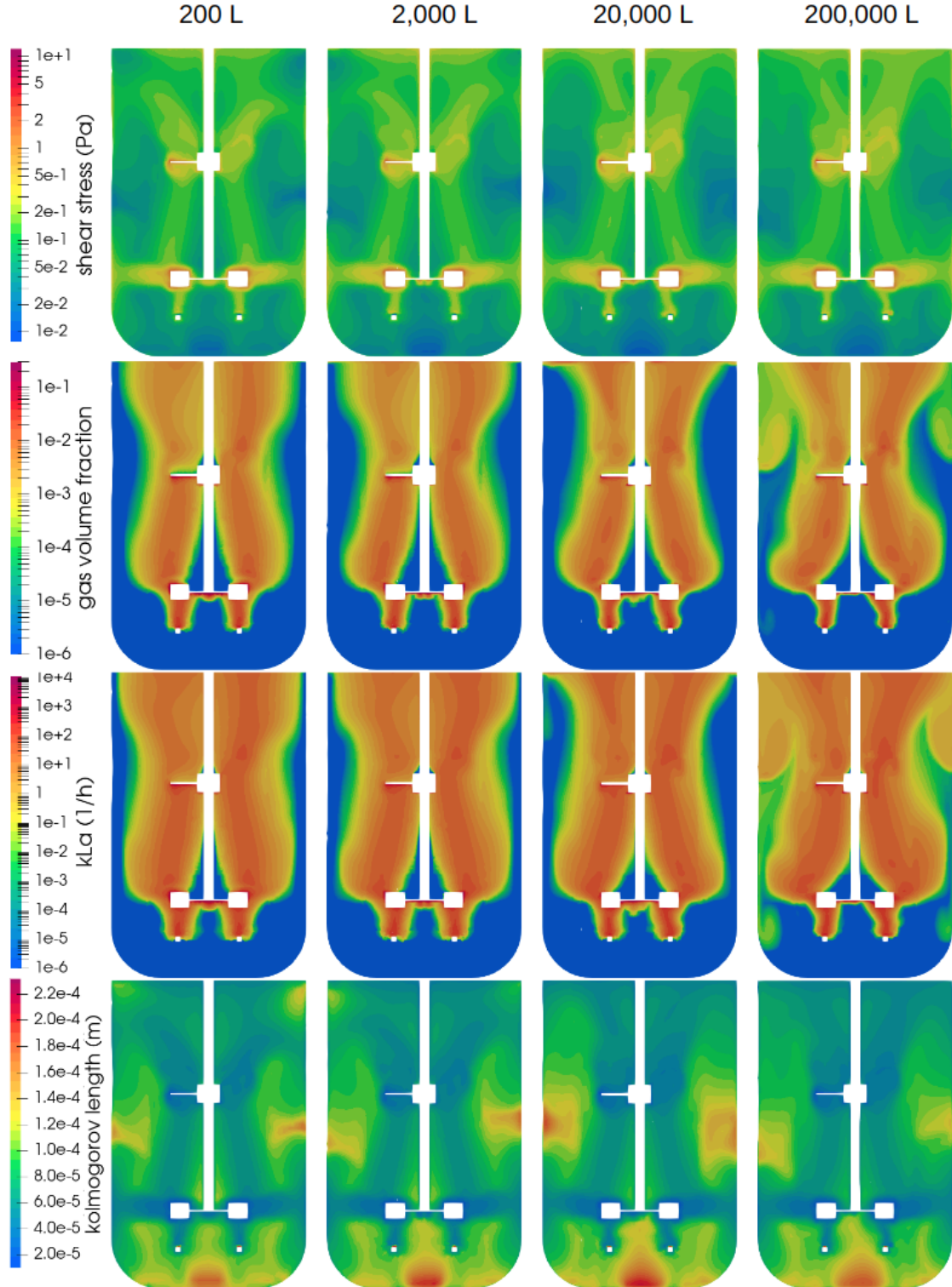


Figure 8: Results of time-averaged field values in a slice along the impeller blades, not in a plane with the baffles. First row: shear stress (Pa); second row: gas holdup (volume fraction); third row: kLa (h^{-1}); fourth row: Kolmogorov length (m)

of $7\text{-}10\text{ h}^{-1}$ to maintain cell health with 10^6 cells/mL. Although oxygen requirements are cell-line specific, higher aeration may be required to maintain cells at higher density. Using a typical oxygen consumption value for CHO cells of $0.32\text{ pmol} / \text{cell h}$ (Xing, Kenty, et al., 2009) and a saturation oxygen concentration of 1.07 mM (Kolev, 2012) corresponding to sparging with pure oxygen, a high cell density of 10^7 cells / mL can be maintained with 40% dissolved oxygen (DO) with a k_La of 5.0 h^{-1} . Zhang et al. (2021) estimated the k_La in two studies that successfully grew bovine cells in spinner flasks (Hanga et al., 2021; Verbruggen et al., 2018), and found values of 4.5 h^{-1} and 1.6 h^{-1} . Hence the k_La predicted here may be sufficiently high to support high density cell culture, but a higher aeration rate might be needed for some cell types.

3.4.3.2 Energy Dissipation and Shear Stress Maximum energy dissipation rates ranged from 10 W/kg to 365 W/kg , in a similar range to what was found in Villiger et al. (2018) across 15 mL to $15,000\text{ L}$ scale, although the maximum values were reached in a negligible volume fraction. While adherent cells relevant to cultivated meat may have different sensitivity, CHO cells have been found to be resilient to very high levels of energy dissipation, $10^4 - 10^5\text{ W/kg}$ (Ma, Koelling, and Jeffrey J. Chalmers, 2002). However, sub-lethal effects may occur at much lower levels. For instance, Sieck et al. (2013) found a decrease in CHO cell monoclonal antibody productivity with exposure to as low as 0.4 W/kg , although this is not a consideration for cultivated meat production. Godoy-Silva, Mollet, and Jeffrey J. Chalmers (2009) found some reduction in CHO cell growth with chronic exposure to 290 W/kg . While technically the $20,000\text{ L}$ case reached that level somewhere, the volume fraction above 290 W/kg was negligible ($<10^{-11}\%$). In fact, in all cases $<0.06\%$ of the volume was above 1 W/kg . While more research is needed to determine thresholds for lethal and sublethal effects on relevant cell lines, the values found here at least do not appear damaging to CHO cells.

Shear stress was calculated from energy dissipation at sub-Kolmogorov scales according to Equation (33), and in all cases averaged from $0.099 - 0.105\text{ Pa}$. Neunstoecklin et al. (2015) found CHO cells were negatively impacted by shear stresses above $32.4 \pm 4.4\text{ Pa}$. The highest shear found in these simulations was $\sim 16\text{ Pa}$, which is well below that threshold. However, as with energy dissipation rate, sublethal effects may occur at lower levels, dependent on cell type. For instance, Park et al. (2010) found that human mesenchymal stem cells showed significant morphological changes at stress

as low as 0.0137 Pa, while L929 (mouse fibroblast) cells showed morphological changes only above 0.05 Pa. Damage to cells from shear and energy dissipation are summarized in Jeffrey J. Chalmers and Ma (2015).

Our simulations do not account for the potentially quite large stress due to bubbles bursting at the surface, which could be a significant source of cell damage (Walls et al., 2017). However, this damage source can be significantly reduced through the use of surfactants (W. Hu, Berdugo, and Jeffrey J. Chalmers, 2011; Ma, Jeffrey J. Chalmers, et al., 2004). In addition, the bursting of bubbles at the gas-liquid interface does not seem to damage cells attached to microcarriers (although cells may be damaged by removal from microcarriers attached to bubbles as they move towards the surface) (Bauer et al., 2000).

3.4.3.3 Kolmogorov Length Turbulent eddies are likely damaging to cells on microcarriers if they are below the critical threshold of $\eta < 0.6d_p$. For CHO cells, a typical cell size is $\sim 15 \mu\text{m}$, and when grown as suspended aggregates, the clusters were found by Han et al. (2006) to reach $\sim 200 \mu\text{m}$ diameter. Similarly, Cytodex 1 microcarriers have an average diameter of $190 \mu\text{m}$, for a critical threshold of $114 \mu\text{m}$.

None of these conditions would cause problems for individual suspended cells with typical diameter $< 20 \mu\text{m}$. However, adipocytes, which will be important for cultivated meat, can reach sizes comparable to microcarriers. As seen in Figure 9, most of the volume at all scales is below the critical threshold for microcarriers, about 94% - 95%. The smallest values are found around the impellers, particularly the Rushton impellers, as well as above the sparger. This is similar to what was found in Zhang et al. (2021), who used two pitched blade impellers, a lower agitation rate, and a higher sparging rate.

3.4.4 Impact of Impeller Type

Pitched blades are more typically used for animal cell culture as they are believed to create less shear than Rushton impellers. However, at scales larger than have been attempted experimentally for animal cells, for instance at 200,000 L, it is important to understand the impact of impeller shape to de-risk fermentor design. Figure 10 shows a comparison between a simulation with one

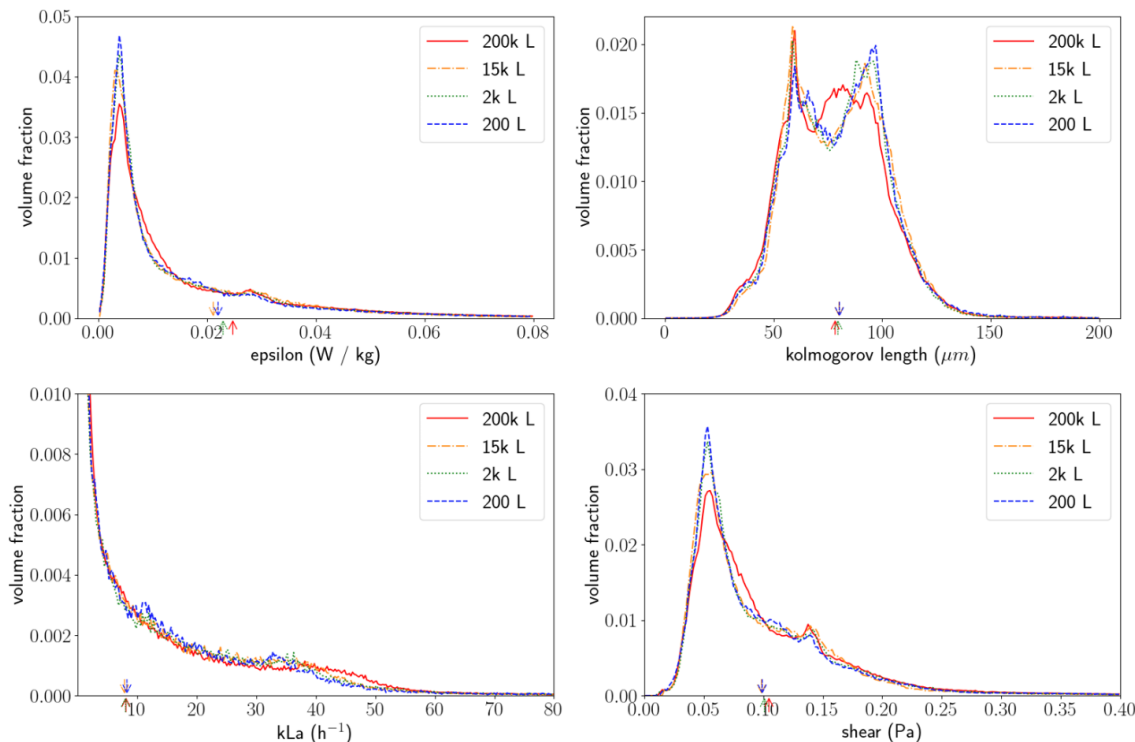


Figure 9: Volume distribution of dissipation, Kolmogorov length scale, $k_L a$, and shear stress. Y-axis is the volume fraction corresponding to each bin. The $k_L a$ graph is cut off below 10^{-4} for clarity since the simulations took on a wide range of values that are approximately 0. The bulk of the volume has ~ 0 $k_L a$. Arrows depict the mean values.

Rushton and one pitched blade impeller versus a simulation with two Rushtons, all at 200,000 L scale with agitation rates calculated to have the same $P/V = 25 \text{ W/m}^3$, according to Equation (31).

The two-Rushton case actually had a higher volume-average $k_L a$ (9.5 h^{-1} vs 8.1 h^{-1}), even though there is slightly less aeration towards the outside of the tank, as seen in Figure 10. Surprisingly, the mean shear is only slightly higher in the two-Rushton case (0.107 Pa vs 0.105 Pa), even though pitched blades are considered lower shear. This is likely because of higher agitation speeds needed with the pitched blade impeller to keep a constant P/V . Figure 11 shows that the volume fractions at the higher end are similar. The similar shear in the two cases may be due to the constant P/V . The two-Rushton case can apparently provide a higher $k_L a$ for a given P/V and shear.

3.5 Discussion

There is little published research currently that considers the hydrodynamics of very large scale bioreactors as it applies to cultivated meat culture. Existing literature on animal cell culture does

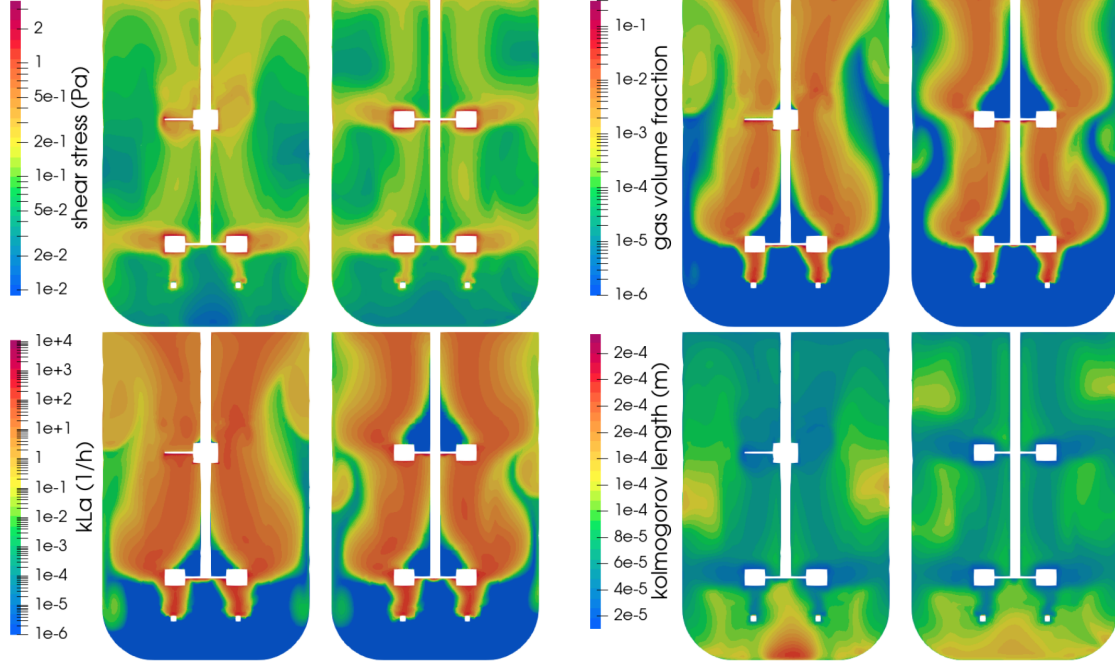


Figure 10: Slices comparing a 200 m³ case with one pitched blade impeller and one Rushton turbine (left) with a 200 m³ case with two Rushton turbines (right).

not examine bioreactors at the 200 m³ scale, but this is likely to be the scale needed for production fermentors in the cultivated meat production process (Negulescu et al., 2023). The results from the scale up study showed that constant P/V and superficial gas velocity enables maintaining similar stress and k_{La} values even up to that scale. Under the conditions considered here, shear stress does not appear likely to be a problem for animal cell culture, which would suggest higher agitation could be used to provide higher mass transfer. However, the Kolmogorov length scale was found to be small enough to pose a threat to microcarrier culture or similarly-sized cell aggregates. In order to increase the Kolmogorov length scale, a lower agitation rate than in these simulations should be used, coupled with a higher aeration rate to maintain sufficient oxygen levels.

The Schiller-Naumann drag model was chosen in order to be conservative, since it predicted the lowest k_{La} of the models tested. However, it may not be the most accurate as it assumes spherical bubbles and is simpler than the Tomiyama and Ishii-Zuber models which depend on the Bond number. The k_{La} predicted in the simulations was potentially too low for high density cell culture, but since shear stress does not appear to be a concern, the aeration could be safely increased to raise the k_{La} . However, these simulations did not consider potential damage due to cells attaching to

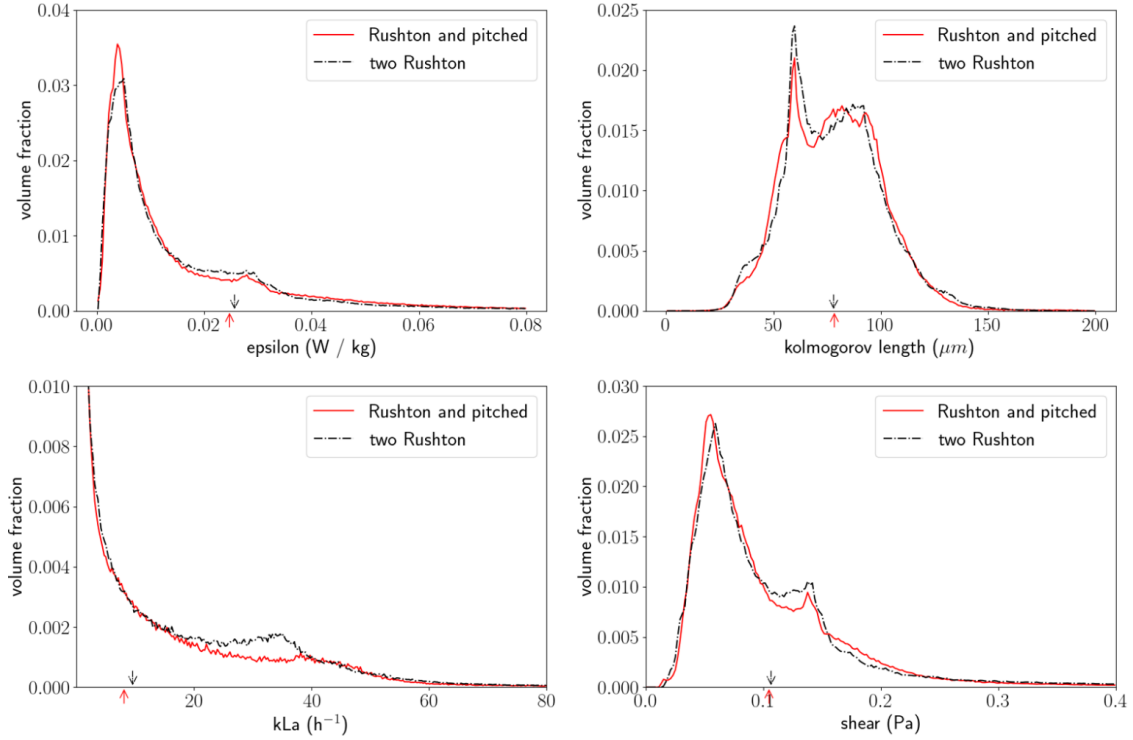


Figure 11: Volume distributions comparing a 200 m³ case with one pitched blade impeller and one Rushton turbine with a 200 m³ case with two Rushton turbines. Arrows depict the mean values.

bubbles and bubbles bursting at the surface, an issue that has been dealt with in biopharma-related animal cell culture by the use of surfactants.

This work relied on some simplified models. The use of RANS turbulence models leads to underestimation of shear stresses, and the use of an Euler-Euler multiphase model means that details of bubble dynamics are not captured. In particular, a single bubble diameter was used, while in a stirred tank there would be a range of sizes due to bubbles being broken up by the impeller and coalescing. As discussed earlier, the choice of bubble diameter likely does not impact the gas distribution much since the terminal velocity will be similar across the relevant diameters (Scargiali et al., 2007), but it could affect the k_La by altering the bubble surface area. In addition, the cell culture media was treated as water and neglected any possible impacts of high cell or microcarrier density on the fluid properties.

The results presented here all use time-averaged values, while a more detailed investigation would consider transient fluctuation, perhaps using large eddy simulation to obtain more detail on the

turbulent eddies. Animal cell culture typically occurs over the course of days, but a transient CFD simulation of such a long span of time is computationally impractical. This could be addressed using CFD-based compartmental models such as Nadal-Rey, Dale D. McClure, Kavanagh, Cassells, et al. (2021) and Nauha, Kálal, et al. (2018), which could be built from CFD results like those presented here. To our knowledge, there are no other CFD simulations in the literature at this scale that use impeller types and operating conditions typical of animal cell culture. Since cultivated meat will likely require culturing animal cells in an order of magnitude larger bioreactors than have been used previously, simulations like these are an important step towards making cultivated meat commercially viable, especially when combined with cell growth models in which oxygen and carbon dioxide concentrations are incorporated as well as the impact of localized shear stresses.

4 Cell Growth Models and Integration with CFD

4.1 Introduction

The first part of this chapter covers work fitting growth models to data collected from mouse myoblast (C2C12) cells. These fits identified the growth factor FGF2 as a potential limiter of the cell growth, motivating further experiments. The second part covers experiments to investigate this possible role of initial FGF2 concentration on cell growth. The third part integrates cell growth models with the CFD models developed in the previous chapter through compartment modeling, with the inclusion of some models of hydrodynamic damage due to turbulent eddies as well as bursting bubbles.

4.2 Modeling of Cell Growth

4.2.1 Background

The Monod model, which has the same form as Michaelis-Menten kinetics, is widely used to describe the relationship between cell growth and substrate concentration. While it is considered an empirical relationship rather than a mechanistic one, it has some physical justification similar to that for Michaelis-Menten enzyme kinetics. Assuming there are a limited number of transport sites, they will all be saturated at sufficiently high nutrient concentrations, meaning the uptake rate does not depend on concentration. But when the nutrient levels are low enough to start leaving transport sites empty, the uptake rate becomes concentration-dependent (Breese and Admassu, [1999](#))

The basic Monod model is expressed as

$$\frac{dX}{dt} = \mu X, \tag{38}$$

$$\mu = \mu_m \frac{[S]}{K_S + [S]}, \tag{39}$$

where X is the cell density, μ is the specific growth rate, μ_m is the maximum specific growth rate, $[S]$ is the substrate concentration, and K_S is the monod constant for substrate S . More complex

variations include a product of multiple limiting terms as well as inhibiting terms. A general form for metabolites S_i and inhibitors I_i might look like (Ben Yahia, Malphettes, and Heinzle, 2021)

$$\mu = \mu_m \prod \frac{[S_i]}{K_i + [S_i]} \prod \frac{KI_i}{KI_i + [I_i]}, \quad (40)$$

where KI_i is an inhibitory Monod constant. Metabolites may generally have both growth-associated and non-growth-associated (maintenance) utilization terms. The specific utilization rates for nutrients can be written as

$$\frac{dS}{dt} = -\frac{\mu X}{Y_{X/S}} - m_S X, \quad (41)$$

where $Y_{X/S}$ is the yield rate of cells per substrate, and m_S represents maintenance consumption not associated with cell growth. Similarly, production of products like lactate can be written as

$$\frac{dP}{dt} = Y_{P/X} \mu X + m_P X, \quad (42)$$

where $Y_{P/X}$ is the yield rate of product per added cell.

These models are favored for their simplicity and effectiveness (Kyriakopoulos et al., 2018). Glucose and glutamine are often used as limiting substrates, while lactic acid produced during glucose metabolism is often used as an inhibitory product. However, each added term increases the number of parameters that must be fit via nonlinear regression.

For adherent cells, growth may be further constrained by confluence. Such cells generally grow in a single layer and cease proliferation once they are surrounded by other cells. As a result, typical nutrient-limited models may no longer apply once the cells density increases beyond a critical value. To capture confluence limitations, combined models incorporating both nutrient and confluence limitations have been developed. Xu (2020), provided analytical solutions to models of this form, using the specific growth rate

$$\mu = \mu_m \frac{S}{K_S + S} \left(1 - \frac{X}{X_m}\right). \quad (43)$$

Jiang et al. (2019) extended this by adding an exponent on the capacity term, allowing for greater

flexibility in modeling the rate at which growth slows near capacity:

$$\mu = \mu_m \left(1 - \left(\frac{C_X}{C_X^{max}} \right)^\alpha \right). \quad (44)$$

However, it is possible to culture adherent cells such that space is not the limiting factor. For example, Breese and Admassu (1999) fitted a Monod model to C2C12 (mouse myoblast) cells grown on microcarriers in flasks. Although confluence eventually limits growth, their experiments captured several cell doublings before confluence was reached, allowing them to identify a glucose-limited Monod relationship. Thus it is possible to isolate nutrient limitations for adherent cell types.

4.2.2 Data

The data for initial model fitting were sourced from O'Neill et al. (2022), focusing on chicken muscle precursor (cMPC) and mouse myoblast (C2C12) cells. Chicken myofibroblasts (cMFB) were excluded since they showed pure exponential growth throughout the recorded growth phase data, making it impossible to model any nutrient-driven change in growth rate. The cell counts were converted to cells/cm² and the metabolites were converted to millimolar (mM) for most terms, except for FGF2 which was expressed in picomolar (pM) to be on a similar numerical scale as the others.

Counts of both cell types declined late in the culture, but it is unclear whether this is due to cell death or differentiation, as differentiation involves fusing of cells into multi-nucleated fibers. Due to this uncertainty, and the fact that in practice cell cultures would not be allowed to grow to that point, simple models without death rates were fitted to the data up to the observed growth peak.

4.2.3 Methods

Models like Equation (41) were fitted to each of glucose and glutamine. However, since FGF2 degrades quickly, an additional degradation coefficient d_F was added to the FGF2 equation

$$\frac{dF}{dt} = -\frac{\mu X}{Y_{X/F}} - m_F X - d_F F. \quad (45)$$

The models considered included limitation due to only glucose or only FGF2, with or without the logistic inhibition term. Models involving more terms were considered as well. A representative example is a model involving glucose and glutamine limitation as well as lactic acid inhibition. The growth rate in this case looks like

$$\mu = \mu_m \frac{[S]}{K_S + [S]} \frac{[G]}{K_G + [G]} \frac{K I_L}{K I_L + [L]}, \quad (46)$$

where $[S]$ is the glucose concentration, $[G]$ is the glutamine concentration, and $[L]$ is the lactic acid concentration.

The fitting process began with parameter estimation directly from the data where possible. The maximum specific growth rate, μ_m , was estimated from the slope of a linear fit to the logarithm of cell counts over time. Specifically, an initial estimate for μ_m was derived from the steepest slope between the first data point and successive points. Yield coefficients were estimated by calculating the change from day 0 for each substance and plotting one of these delta sets against the other (e.g., cell deltas vs glucose deltas to estimate $Y_{X/S}$). The slope of a linear fit to this plot was used as the initial yield rate estimate. Finally, the maximum cell density, X_m , was estimated as the maximum observed cell density.

These values were used as starting points for the fitting. In addition, the glucose Monod constant K_S was initially estimated at 2.5 mM, based on Breese and Admassu (1999) who found this value for C2C12 cells. For fitting, all the parameters were constrained to range from 20% to 500% of their initial estimate. Limiting factors included glucose, glutamine and FGF2, while lactate acted as an inhibiting term. Models incorporating different combinations of these terms, alongside logistic growth, were fitted using BFGS, with `Fminbox` from the Julia Optim library to enforce the bounds on each parameter.

To validate the fitting process, simulated data were first generated using the different proposed models with the addition of Gaussian noise to each field with magnitude $\sim 1 - 10\%$ of the field's maximum value. Fitting the simulated data confirmed the method's ability to recover the original parameters. Subsequently, this approach was applied to the experimental data.

As mentioned above, it is expected that a logistic equation should describe the growth of these

adherent cells well, at least if they are allowed to grow to confluence. However, it is possible that including other terms could enhance the fit if they reached limiting levels before the cell growth was stopped due to confluence. It is not straightforward to compare models that are fitting different numbers of metabolites along with the cell counts. To enable direct comparison between models with different inputs, analytic curves were fit to the nutrients. These were all generalized logistic equations (GLE), as in Goudar et al. (2005), except for FGF2 which was better fit with a simple exponential decay. These analytical curves were used as input to versions of the models that fit only cell growth based on these nutrient levels.

This enables comparison via Akaike Information Criterion (AIC), which measures the model fit error with a penalty for the number of parameters. The models were fit with least squares regression, so the form of the AIC is given by (Portet, 2020)

$$AIC = 2k + n \ln SSE, \quad (47)$$

where k is the number of parameters, n is the number of data points, and SSE is the sum of squared errors. AIC may give misleading results for a relatively small number of data points. In this case a corrected version, AICc, can be used (Portet, 2020):

$$AICc = AIC + \frac{2k^2 + 2k}{n - k - 1}. \quad (48)$$

Only the differences in the AIC values matter, with the lowest AIC indicating the preferred model, balancing fit quality with number of parameters.

4.2.4 Results

Figure 12 illustrates model fits to the cMPC and C2C12 data, where the cell growth and the metabolites were simultaneously solved and fitted. The fits are qualitatively reasonable in all cases, although the glucose-limited model does not capture the data as well, particularly for the cMPC cells, as it fails to match the decline in cell growth towards the end.

To better understand whether the nutrient limiting terms provide information beyond the logistic model, cell growth-only models were fitted based on analytic curves approximating the glucose and

FGF2. Figure 13 shows the results of these fits, and Tables 6 and 7 summarize the fitted parameters. AIC and AICc values (Tables 8 and 9) indicate that logistic and FGF2-only models provided the best fits. However, for cMPC cells, the FGF2 model performed slightly better, while for C2C12 cells, the logistic model was superior.

In all cases, incorporating a nutrient limitation into the logistic model did not improve the AIC score, even if it slightly lowered the error (Tables 8, 9). The improvement in fit did not justify the increased number of parameters. For glucose-limited models, the glucose Monod constant K_S reached the upper bound on the parameter fit. In the glucose-logistic model, this means the fit was as close as it could get to the pure logistic model. In the glucose-only model, this just indicates that it is approaching an exponential fit with no influence from the glucose. As a result, the glucose-limited model does not actually capture the decline in growth rate towards the end of the time period.

Table 6: Parameters obtained for several models fitted to the cell growth only. cMPC cells.

Parameter	Logistic	Glucose	Glucose Logistic	FGF2	FGF2 logistic
$\mu_m (d^{-1})$	0.631	0.317	1.168	1.295	1.155
$X_m (\frac{\text{cells}}{\text{mm}^2})$	1997.99		2013.369		10305.556
$K_S (mM)$		12.5	12.5		
$K_F (pM)$				2.804	2.053

Table 7: Parameters obtained for several models fitting the cell growth only. C2C12 cells.

Parameter	Logistic	Glucose	Glucose Logistic	FGF2	FGF2 logistic
$\mu_m (d^{-1})$	0.905	0.822	0.915	1.023	0.915
$X_m (\frac{\text{cells}}{\text{mm}^2})$	3486.913		3492.446		3743.043
$K_S (mM)$		12.5	0.125		
$K_F (pM)$				0.767	0.05

Table 8: AIC and corrected AICc for several models fitted to the cMPC cell growth data, as well as the sum squared error for each model compared to the data.

Model	AIC	AICc	SSE
Logistic	31.65	33.65	21.59
Glucose	42.45	44.45	71.71
Glucose Logistic	33.51	38.31	21.25
FGF2	30.71	32.71	19.45
FGF2 logistic	32.76	37.56	19.55

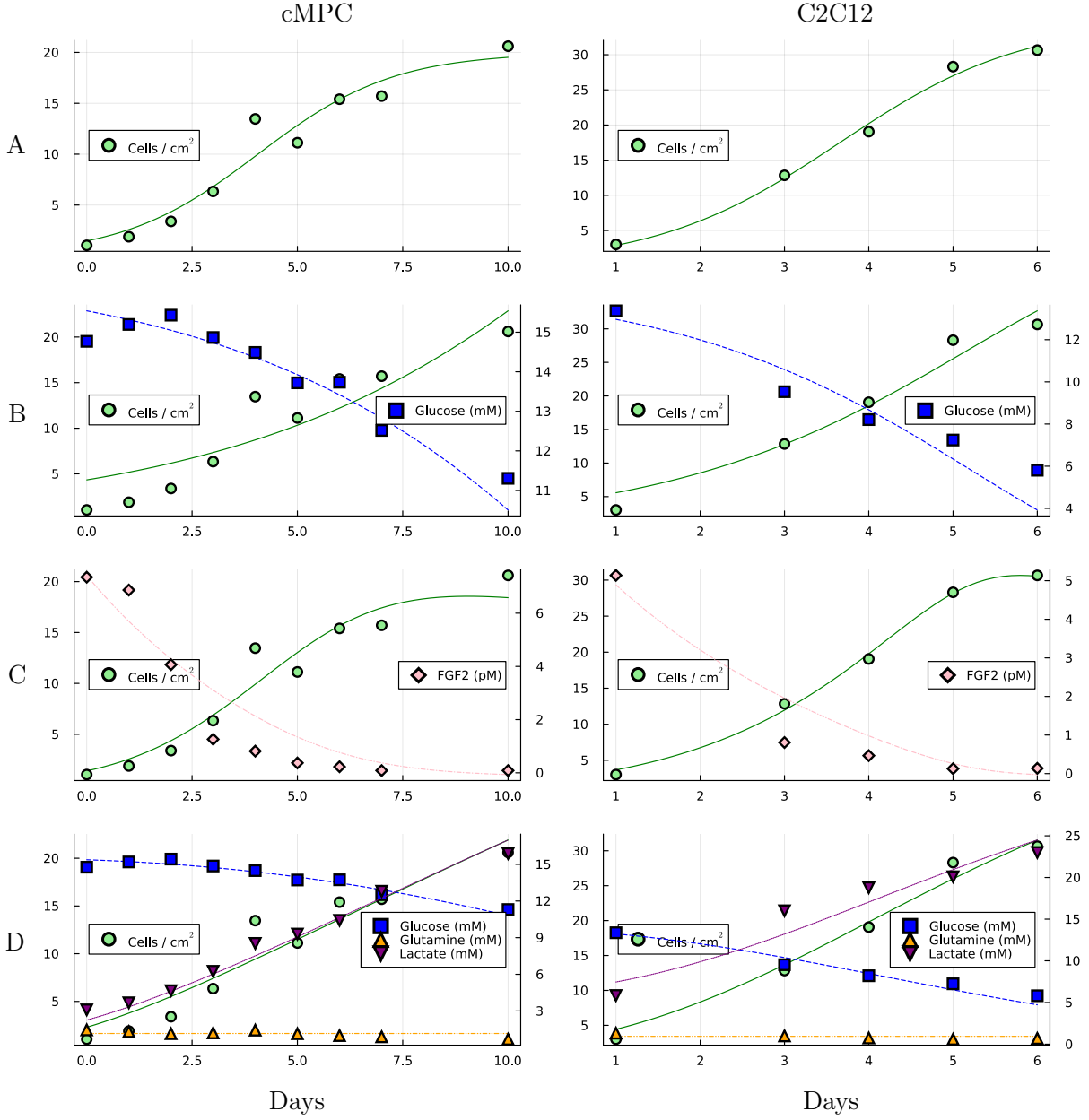


Figure 12: Example fits of several limiting terms as well as a model combining several limiting terms for cMPC (left) and C2C12 (right) cell data. A) logistic model; B) glucose-limited model; C) FGF2-limited model; D) model limited by glucose and glutamine, inhibited by lactate.

4.2.5 Discussion

The logistic and FGF2-limited models outperformed the others in terms of AIC, suggesting that growth was limited either by confluence or by FGF2. As expected, including glucose- and glutamine-limitation did not improve the fit. Since neither nutrient was close to running out by the time the cell growth ceased, they were not good candidates for a limiting factor. In contrast, FGF2 does run

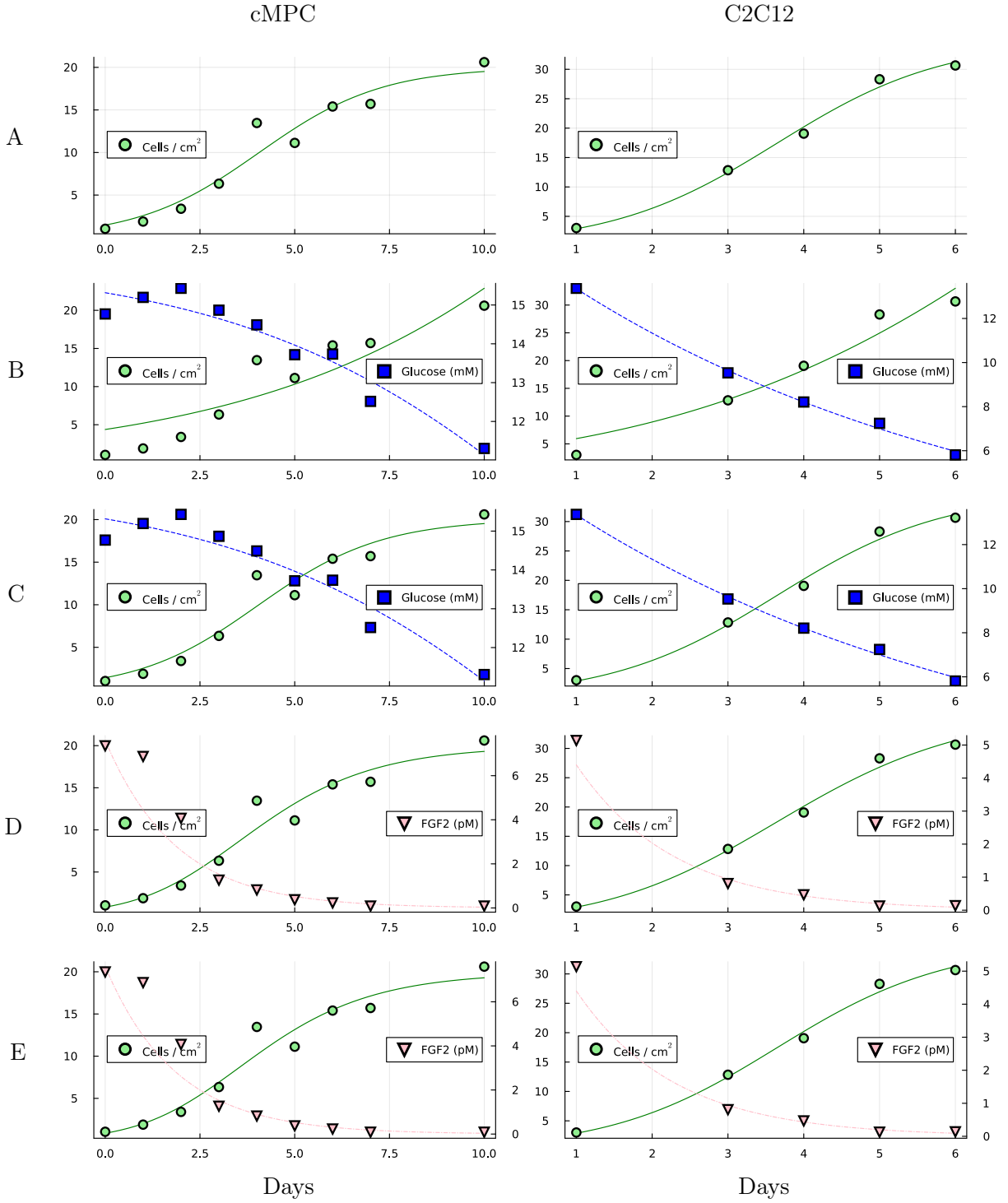


Figure 13: Results of several models fit to the growth phase of the cMPC (left) and C2C12 (right) cell data, using a logistic curve fit to the glucose data as input for models that depend on glucose. A) logistic only; B) glucose only; C) glucose-logistic; D) FGF2 only; E) FGF2-logistic.

Table 9: AIC and corrected AICc for several models fitted to the C2C12 cell growth data, as well as the sum squared error for each model compared to the data.

Model	AIC	AICc	SSE
Logistic	10.29	16.29	3.52
Glucose	20.30	26.30	26.07
Glucose Logistic	12.31	36.31	3.53
FGF2	10.84	16.84	3.93
FGF2 logistic	12.42	36.42	3.61

out around the time the cell growth stops. More investigation is necessary to determine whether the FGF2 is in fact the limiter, or if it is just confluence that stops the proliferation.

These results indicate that under the experimental setup of adherent cells grown in well plates in PCGM media, growth may reach confluence before nutrient depletion becomes a significant factor. Therefore this dataset alone cannot provide sufficient insight into nutrient-limited growth dynamics. Future studies could potentially resolve this problem by using media with lower initial nutrient concentrations, or by using microcarrier culture to provide more surface area relative to media and thus avoid confluence long enough to isolate nutrient limitations.

4.3 FGF2 Experiments

4.3.1 Background

Fibroblast growth factor 2 (FGF2) is a signaling protein involved in controlling cell growth. Growth factors like FGF2 currently make up a large portion of the cost of culture media (Specht, 2020), making their role and optimal use an important area of study.

As shown in the previous section, a naïve fit of a Monod relationship between this specific rate and the data on glucose demonstrates the lack of a Monod relationship. This is unsurprising, because nutrients like glucose and glutamine simply did not decrease enough by the time the cells stopped growing. However, applying this same method to FGF2 levels revealed a strong fit. This is expected, as the FGF2 shows a clear exponential decay, so the sigmoidal pattern of the cell growth is consistent with a Monod relationship.

The decay of the FGF2 was most likely due to degradation (Ding and Peterson, 2021) rather than consumption by cells, making a Monod relationship unlikely *a priori*. However, the main function

of FGF2 is to prevent differentiation of cells by blocking myogenin. Therefore, FGF2 may act more like a threshold-dependent signal in which growth is unaffected at high FGF2 levels when all the myogenin is being blocked, but declines significantly when FGF2 levels drop below a critical level. Additional experiments were conducted by growing cells at varying initial FGF2 concentrations to test the relationship between cell growth and FGF2.

4.3.2 Methods

C2C12 cells were cultured in both serum-containing media (PCGM) and serum-free media (Essential 8 base supplemented with TGF- β to match standard Essential 8 levels), with varying levels of FGF2 added. The PCGM (Prime Complete Growth Medium) consists of 40% DMEM, 40% F10, and 20% fetal bovine serum (FBS). The cells were seeded in 12-well plates, with 3 replicates at each FGF2 level.

For serum conditions, initial FGF2 levels were set at 0 ng/mL, 100 ng/mL, and 200 ng/mL. The serum-free conditions included these levels and an additional 2.5 ng/mL level. Cells were counted using Hoechst stain and fluorescent imaging with the ImageXpress Pico and CellularReporterXpress software.

4.3.3 Results

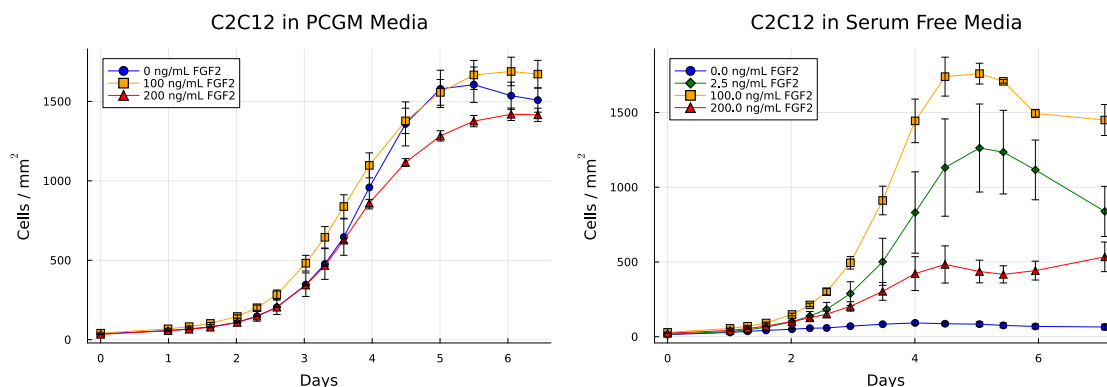


Figure 14: Data collected for C2C12 cells grown with different starting levels of FGF2 in PCGM media (left) and serum free media (right).

Cell density outcomes varied with initial FGF2 concentration (Figure 14). In serum-containing media, final cell densities were similar across 0 ng/mL and 100 ng/mL FGF2 levels, but the 200

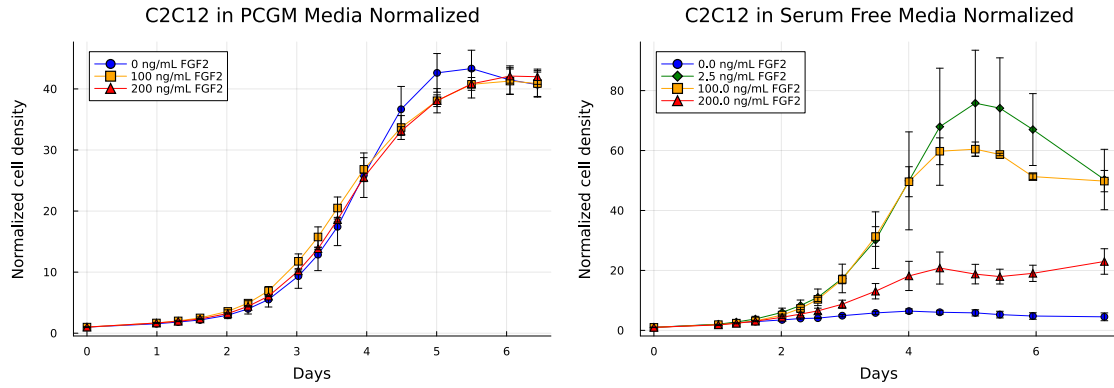


Figure 15: Data collected for C2C12 cells grown with different starting levels of FGF2 in PCGM media (left) and serum free media (right), normalized by starting density.

ng/mL condition resulted in a slightly lower density. However, normalizing for initial seeding density reduced this difference (Figure 15).

In the serum-free media, cell growth was highly dependent on initial FGF2 concentration. Cells started with no FGF2 showed minimal growth, and the final cell density increased as starting FGF2 increased up to 200 ng/mL, where growth was again inhibited.

Applying one-way ANOVA with Tukey's post-hoc test shows that the maximum cell densities across all initial FGF2 levels are significantly different within the serum-free media ($p < 0.05$). In the serum-containing media, there is no significant difference between 0 ng/mL and 100 ng/mL, but there is between 100 ng/mL and 200 ng/mL ($p=0.02$).

4.3.4 Discussion

The results reveal distinct FGF2 effects depending on the media type. In serum-containing media, cells grew well without added FGF2, suggesting that serum provides alternative factors that replace FGF2's role. However, very high FGF2 concentrations (200 ng/mL) inhibited growth, a trend that lacks a clear mechanistic explanation.

In serum-free media, FGF2 was essential for growth, with increasing FGF2 concentrations generally promoting higher final densities. However, the highest FGF2 level (200 ng/mL) again led to growth inhibition, even more pronounced than in the serum-containing media.

It is worth noting that, in the serum-free media, cell growth with an initial FGF2 level of 2.5 ng/mL

was comparable to that at 100 ng/mL, which is the standard amount in Essential 8. This suggests that a significantly lower FGF2 level is sufficient for cell growth, which could help to reduce media costs.

Several factors may have influenced these findings. The cell counts at high density are unreliable, as the automated counter struggles to distinguish nuclei when the cells are densely packed without also detecting non-existent cells in the background noise. In addition, the average starting cell counts were slightly different in the different cases, which would at least have an effect of delaying the time at which cells reach a given density. If the loss of FGF2 is indeed primarily due to degradation independent of the cells, it will run out at a fixed time. Thus, cells that started at a lower density would still be at a lower density when the FGF2 runs out. This implies that if FGF2 is preventing cells from differentiating, they may start differentiating at a lower density.

Similarly, cultures started at a lower density might reach a lower maximum density if there is anything reducing the cell growth rate over time. The frequent imaging with Hoechst stain could be such a slowing factor (Purschke et al., 2010). This imaging could affect the final density reached for cells that started at a lower density, if for instance cells cease growth after a given number of exposures to the fluorescent imaging.

The method of counting cells via Hoechst stained nuclei is also different from what was used in the original spent media analysis (O'Neill et al., 2022), where cell counts were obtained manually via hemocytometer. When muscle cells differentiate, they fuse into multi-nucleated fibers. Hence the nuclei counts would not show a decline due to differentiation, while the hemocytometer counts would. However, that does not explain why the cells reached a significantly lower final density in these experiments. This difference could be due to this particular batch of cells being different, as they were a high passage number, or it could potentially be due to the frequent fluorescent imaging.

These findings suggest that FGF2 is crucial for cells grown in serum-free media, and that it is important to find the right level to avoid inhibiting growth. However, the necessary level may be significantly lower than what is in commercial Essential 8 media. Future experiments should avoid repeated fluorescent imaging and use lower passage cells to validate these results.

4.4 Integration of Cell Growth and CFD Models

4.4.1 Background

Chapter 3 presented CFD simulations of geometrically similar bioreactors ranging from 200 L to 200,000 L. By analyzing the static distribution of stresses, it was possible to evaluate whether the hydrodynamic conditions in these reactors were broadly suitable for cell growth. However, quantifying the specific effects of these conditions on cell proliferation requires integrating cell growth models with the CFD solutions. This section demonstrates a compartment modeling approach, combining cell growth and hydrodynamic damage models with the CFD data to predict growth dynamics.

4.4.2 Methods

4.4.2.1 Compartment Model Construction The compartments were built from the CFD results, and were defined as nested annular cylinders, as illustrated in Figure 16. This was done by a custom Python script which reads a configuration file specifying the locations of vertical and radial cuts. This script generates a dictionary for OpenFOAM's `topoSet` utility, which defines sets of cells in the mesh. The script also creates function objects for each of these cell sets to calculate the field averages required by the compartment model. An additional script calculates fluxes between compartments in the open source post-processing software ParaView.

The resulting data files provide values for each compartment. These were input into a separate script that solves a cell growth model. This script provides a general growth model solver, which takes as input a function describing the cell growth as well as optional functions that calculate specific death rates from damage based on the local hydrodynamic field values, such as a model of eddy damage or a model of bubble bursting damage.

4.4.2.2 Cell Growth Model As there are not currently any detailed growth models for cultivated meat-relevant cells, the integration here was done with a model based on one from the literature developed for CHO cells. Similar to the model from Farzan and Ierapetritou (2018), the model here is a slightly simplified version of the model in Xing, Bishop, et al. (2010).

This model defines the specific growth rate by incorporating limiting terms based on glucose and



Figure 16: An illustration of compartments constructed using the OpenFOAM utility `topoSet`.

glutamine and inhibiting terms based on lactate and ammonia:

$$\mu = \mu_{\max} \frac{[Glc]}{K_{Glc} + [Glc]} \frac{[Gln]}{K_{Gln} + [Gln]} \frac{KI_{Lac}}{KI_{Lac} + [Lac]} \frac{KI_{Amm}}{KI_{Amm} + [Amm]}. \quad (49)$$

The specific death rate is similarly based on lactate and ammonia:

$$\mu_d = \mu_{dm} \frac{[Lac]}{KD_{Lac} + [Lac]} \frac{[Amm]}{KD_{Amm} + [Amm]}. \quad (50)$$

Then the viable cell density is given by

$$\frac{dX}{dt} = (\mu - \mu_d)X. \quad (51)$$

The other metabolites are modeled as in the following equations, with yield terms of the form $Y_{A/B}$, maintenance usage terms m_A and degradation terms d_A :

$$\frac{d[Glc]}{dt} = - \left(\frac{\mu}{Y_{X/Glc}} + m_{Glc} \right) X, \quad (52)$$

$$\frac{d[Gln]}{dt} = - \left(\frac{\mu}{Y_{X/Gln}} + m_{Gln} \right) X - d_{Gln}[Gln], \quad (53)$$

$$\frac{d[Lac]}{dt} = Y_{Lac/Glc} \left(\frac{\mu}{Y_{X/Glc}} + m_{Glc} \right) X, \quad (54)$$

$$\frac{d[Amm]}{dt} = Y_{Amm/Gln} \left(\frac{\mu}{Y_{X/Gln}} \right) X - r_{Amm} X + d_{Gln}[Gln]. \quad (55)$$

Also like Farzan and Ierapetritou (2018), this model included an equation for dissolved oxygen,

$$\frac{d[DO]}{dt} = k_L a (DO_{eq} - [DO]) - X \cdot OUR, \quad (56)$$

with a specific oxygen uptake rate of 3.2×10^{-10} mmol cell⁻¹h⁻¹ (Xing, Kenty, et al., 2009). However, currently this model does not incorporate any impact on cell growth due to oxygen levels.

Here $k_L a$ was calculated based on Higbie's penetration theory as in Equation (7).

The initial cell density was set to 0.2×10^6 cells/mL, as in Xing, Bishop, et al. (2010). The initial concentrations of glucose and glutamine were taken to be 100 mM and 10 mM, respectively, as in Farzan and Ierapetritou (2018), although these values are considerably higher than those used in O'Neill et al. (2022).

4.4.2.3 Eddy Damage Model The model from Lakhotia and Papoutsakis (1992) was used to quantify eddy damage. The idea of this model is that, although there are multiple mechanisms by which turbulent eddies damage cells on microcarriers, all the mechanisms relate to turbulent energy. As a result, they calculated the total turbulent energy in the range of wavenumbers corresponding to eddy sizes that would be damaging to cells. This was calculated by integrating the Pao-Corsin turbulent energy density function:

$$I = \int_{k_c}^{\infty} 1.7 \varepsilon^{2/3} k^{-5/3} \exp \left[-2.55 \nu \varepsilon^{-1/3} k^{4/3} \right] dk, \quad (57)$$

where ε is the turbulent energy dissipation, ν is the kinematic viscosity, and k_c is the wavenumber corresponding to the maximum damaging eddy size and was numerically fitted to cell growth data. The maximum damaging eddy size should be approximately 0.6 times the microcarrier diameter, based on Croughan, Hamel, and D. I. C. Wang (1987).

They used this energy to define a death rate:

$$q = \begin{cases} BI & I > E_0 \\ 0 & I \leq E_0 \end{cases}, \quad (58)$$

where B is also numerically fitted to cell growth data. The threshold E_0 , below which there is no cell damage, was estimated experimentally by growing cells at different agitation rates and finding the approximate energy below which there was no cell death.

4.4.2.4 Bubble Damage Models Two bubble damage models were implemented. The first was constructed based on data from Walls et al. (2017), who numerically simulated bubbles bursting and calculated the liquid volume around the bursting bubble that reached energy dissipation thresholds likely to be deadly to CHO cells. They calculated this volume as a fraction of the bubble volume, and repeated this for a range of bubble diameters. Using their data points showing this ratio for different bubble diameters, a power law between this damaging volume ratio and the bubble radius was extracted (Figure 17).

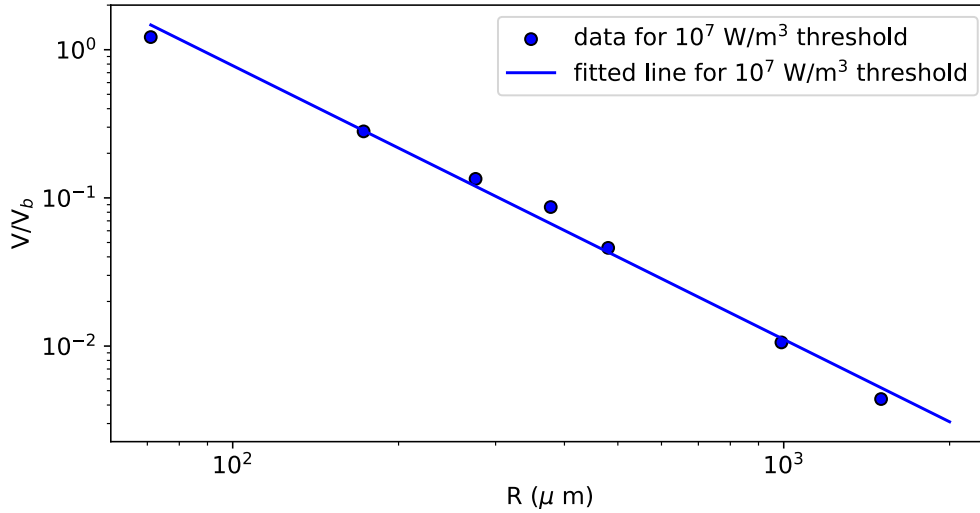


Figure 17: Data extracted from Walls et al. (2017) showing how the damaging volume during bubble burst (V) divided by the bubble volume (V_b) varies with the bubble radius (R). Showing the data using a threshold of 10^7 W/m^3 , along with a best-fit line.

Using this equation, a death rate was calculated only for the top compartments in the bioreactor.

The damaging volume per bubble was obtained by multiplying the volume of each bubble (using the assumed 4 mm bubble diameter) by the result of this fitted equation. The gas flux through the surface of each top compartment was extracted from the compartment data and converted to bubble number flux using the assumed volume for each bubble. Then the death rate is

$$q_{\text{Walls}} = \frac{(\text{death fraction})(\# \text{ bubbles} / \text{time}) \cdot (\text{damaging volume} / \text{bubble})}{\text{compartment volume}}, \quad (59)$$

where the death fraction is the fraction of cells that are actually killed on exposure to the threshold shear stress. For this work, a value of 0.4 was used for the 10^7 W/m^3 threshold, based on interpolation of data from Godoy-Silva, Mollet, and Jeffrey J. Chalmers (2009).

For comparison, a second bubble damage model was adapted from Cherry and Hulle (1992). Their model assumes that the cells killed by bubbles are some fraction of those in the thin film around the bubble at the surface of the liquid before bursting, and has a death rate of the form

$$q_{\text{Cherry}} = \Psi \alpha \frac{3hF}{rV}, \quad (60)$$

where Ψ is the fraction of cells in the bubble film that die on rupture, α is the ratio of cell density in the bubble film to that in the bulk, h is the film thickness, F is the sparging rate, r is the bubble radius, and V is the total liquid volume. They experimentally determined the death fraction Ψ by dipping a wire loop in medium containing suspended cells to form a thin film and counting the surviving cells after the film spontaneously ruptured compared to those collected from the film without rupture. They determined the ratio of cell density in the film to that in the bulk by comparing the number of cells in the bulk as an increasing fraction of the volume became foam. These experiments were done with Sf-9 insect cells, and they found the values to be $\alpha \approx 0.6$ and $\Psi \approx 0.2$. To estimate the film thickness h , they used the average cell diameter $\sim 15 \mu\text{m}$.

For use in the compartment model, the sparging rate F was replaced with the gas flux through the top of each compartment, and the volume V was taken to be the compartment volume. As in the previous model, this death rate was only calculated for the top compartments, and set to 0 in all other compartments. The values of α , Ψ and h from Cherry and Hulle (1992) were used.

4.4.3 Data

The CFD data used for the compartment modeling are the results from the previous chapter, in which geometrically similar bioreactors with one Rushton turbine and one pitched blade impeller were simulated at the 200 L, 2,000 L, 20,000 L, and 200,000 L scales, keeping constant P/V and superficial gas velocity. Those simulations were averaged over 30 seconds starting after the solution was sufficiently stable, and these time-averaged values were used to construct the compartment models.

4.4.4 Results

Figure 18 shows the results of a simulated batch process solved in the compartment model. Since there is no hydrodynamic damage included here, the results look the same at all the scales. The cells decline eventually due to the glutamine running out. The equilibrium oxygen concentration was taken to be that corresponding to sparging with pure oxygen. The k_La values in the CFD simulations at all scales were $\sim 8 \text{ h}^{-1}$. The cell density only reached $< 2.5 \times 10^6 \text{ cells/mL}$ under the given batch conditions. However, a k_La of 8 h^{-1} can support up to 10^7 cells/mL with the specific oxygen uptake rate of $3.2 \times 10^{-10} \text{ mmol cell}^{-1}\text{h}^{-1}$.

The results of the eddy damage model show a significant impact of incorporating eddy damage at all. In contrast, there is little difference between the scales apart from a slight trend towards more damage at the larger scales. These results are not surprising, since the previous chapter found that the distribution of Kolmogorov length scale was similar across the scales, and was small enough to be damaging to cells on microcarriers in nearly all of the liquid volume.

Both bubble models predicted significant cell death, with trends showing reduced damage at larger scales. This aligns with the fact that the aeration was scaled keeping superficial gas velocity constant, meaning that the volumetric gas flow rate is proportionally lower at the larger scales. It is also true in general that the surface area proportionately decreases as the volume increases, which could lead to a similar trend even with constant volumetric flow rate.

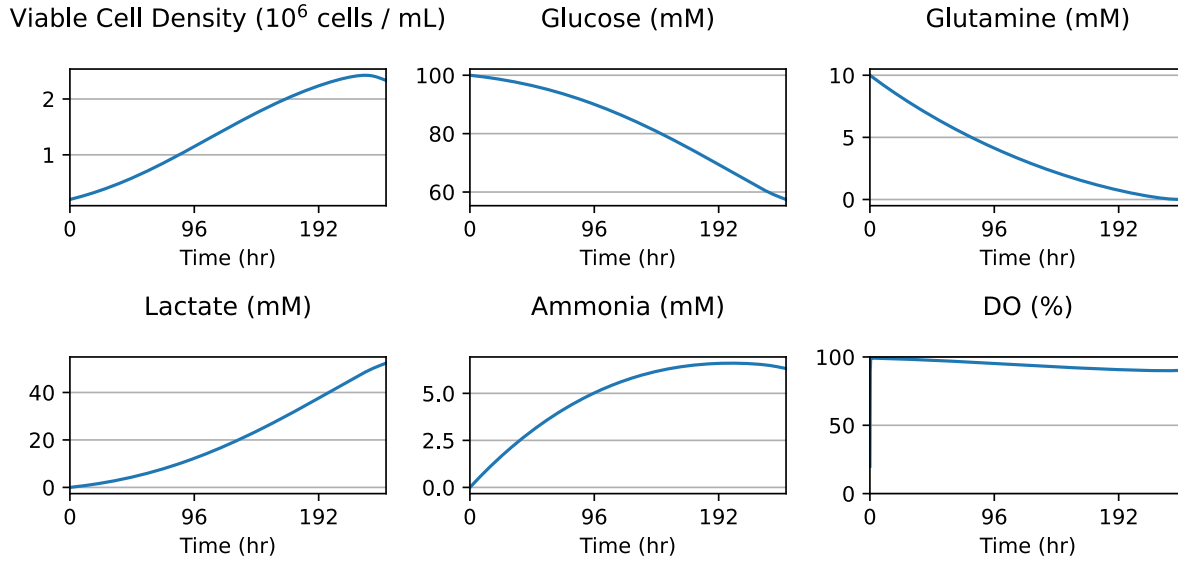


Figure 18: Plots of a simulated batch process in a 200,000 L bioreactor using the cell growth model described above and no hydrodynamic damage models.

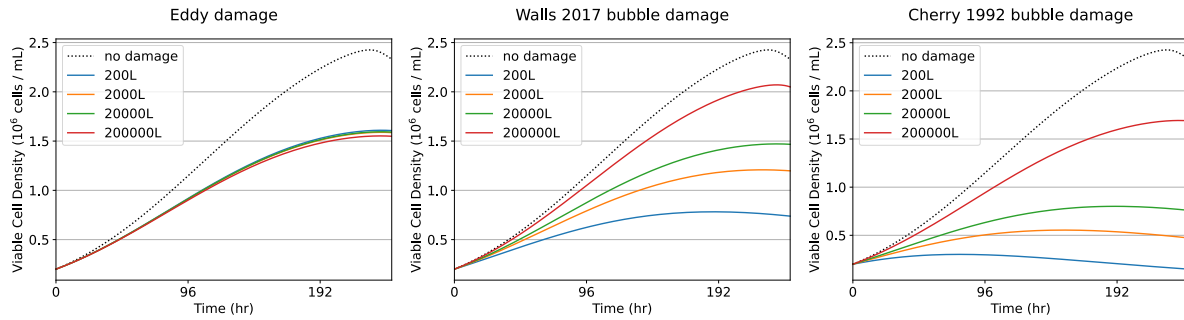


Figure 19: Comparison of baseline cell density over a simulated batch process with no hydrodynamic damage (dotted black line, same at all scales) with the results of incorporating different hydrodynamic damage models at each simulated scale from 200 L to 200,000L. left: incorporation of eddy damage model; center: incorporation of bubble bursting damage model based on Walls et al. (2017) with 10^7 W/m³ threshold; right: incorporation of bubble bursting damage model based on Cherry and Hulle (1992)

4.4.5 Discussion

The results highlight the importance of integrating hydrodynamic damage models into cell growth predictions. Under these models, both eddy damage and bubble-related damage could significantly hinder growth, with bubble damage appearing as a particularly significant concern.

Interestingly, the two bubble damage models produced fairly similar results, despite being mechan-

ically different and being based on different cell types. However, both predicted damage levels that seem excessively high, especially given the low aeration rate used in the CFD simulations. These results imply that it would be virtually impossible to grow cells at the 200 L scale, which is not accurate. This suggests that the models could benefit from some refinement.

One limitation stems from the compartment model's assumption of uniform cell density within each compartment. In reality, cells may be less dense near the surface of the tank, especially following localized cell death due to bubble bursting. It should take some time for the cell density at the surface to be replenished, but this is not captured by the current model. This could explain why reducing the vertical size of the top compartments reduced the cell death rate. The results shown in Figure 19 (center) were obtained by making the top compartments as thin as possible, being only one computational cell thick.

Additionally, the high death rate around bursting bubbles used in the model likely contributes to the overestimation of bubble damage. While Cherry and Hulle (1992) were considering cells only in the thin film above a bursting bubble, even there they found that only about 20% of the cells were killed when the bubble burst. The death rate of 40% taken from Godoy-Silva, Mollet, and Jeffrey J. Chalmers (2009) was conservatively high, as their death rates were significantly higher than the other values from the literature to which they compared their results. Thus, reducing the death rate in the vicinity of bursting bubble could improve the accuracy of the model.

5 Conclusion

This thesis provides insights into the challenges of scaling up bioreactors for cultivated meat production by integrating CFD modeling and cell growth simulations. Chapter 3 demonstrates the feasibility of maintaining consistent hydrodynamic conditions across scales from 200 L to 200,000 L, which is crucial to ensuring cell viability and productivity at commercial scales. By maintaining constant P/V and superficial gas velocity, similar distributions of key parameters like shear stress and k_La were achieved across scales. In addition, using two Rushton turbines instead of one Rushton and one pitched-blade impeller improved the k_La while only slightly increasing the mean shear stress.

However, the analysis highlights specific risks, particularly the potentially damaging Kolmogorov length scales for cells grown on microcarriers or in aggregates. Reducing agitation rates could mitigate this issue, but at the cost of lower mass transfer. Increasing aeration rates may compensate for reduced mixing, but this introduces more bubble-related damage, including shear at the sparger inlet and bubbles bursting at the surface. These findings illustrate the delicate balance between sufficient mixing and cell protection. As demonstrated in Section 4.4, the damage due to these turbulent eddies and bursting bubbles could potentially be highly significant.

Experimental data and growth models for particular animal muscle cells showed that, while the growth factor FGF2 influences cell proliferation, its effects are highly dependent on the background media composition. Excessively high or low levels of FGF2 inhibit growth, whereas moderate levels are not limiting under the tested conditions. These insights guide the optimization of growth media for cultivated meat production.

This work provides a framework to de-risk large-scale bioreactor operations, potentially lowering costs and accelerating the adoption of sustainable meat production technologies. Building on these findings, there are several promising avenues for future research.

Future studies could incorporate more dynamic growth models to better capture shifts in cell metabolism and nutrient use. To be even more accurate, a sophisticated model could be developed, taking into account processes within each cell. This would maximize flexibility to evaluate how the cells will perform in novel environments. However, such models are difficult and costly to

develop, and would have to be done for the particular cell line of interest. Since cell lines are still being developed for cultivated meat, it would not make sense to do that level of characterization currently. However, developing inexpensive Monod models for potentially relevant cell types will still be useful to make better predictions. Similarly, collecting data to quantify the effects of eddy, bubble and shear stress damage on cell types of interest will improve the models developed here.

Additionally, CO₂ accumulation and its effects on cell growth should be integrated into simulations. At large scales, the pressure is significantly higher towards the bottom of the tank, increasing the solubility of both O₂ and CO₂. If that turns out to have a significant impact on cell growth, these models could inform the optimal aspect ratio of large scale bioreactors, since shorter, wider tanks would not reach such high pressures.

Another avenue that would be valuable to explore is cell lifeline analysis. Chapter 3 calculated volume distributions to estimate how much time cells spend in high-stress regions. A more detailed approach, tracking individual particle paths, would provide granular insights into the actual stress histories of cells. Recent advancements in this area, such as those by Haringa, Tang, and Noorman (2022) and Nadal-Rey, Kavanagh, et al. (2023), offer promising methodologies.

While this work focused on stirred tank bioreactors, as discussed in Section 2.4, other types of bioreactors such as airlift and hollow fiber bioreactors may offer unique advantages for cultivated meat production. The framework in Chapter 3 can be readily adapted to airlift bioreactors. However, hollow fiber bioreactors would require more significant changes. Existing literature on computational simulations of hollow fiber bioreactors, such as Menshutina et al. (2020) and Ye et al. (2006), provide a good starting point.

Microcarrier cultures involve complex dynamics, including bead-to-bead transfer and microcarrier aggregation. Future work should incorporate these mechanics to better predict cell distribution and growth. This would extend the current models to include phenomena such as aggregate formation and transfer kinetics, as in Jossen, D. Eibl, and R. Eibl (2021).

By addressing these areas, future research can refine bioreactor design and operation for cultivated meat production. This work underscores the potential of cultivated meat to transform global food systems, offering a sustainable and ethical alternative to traditional meat production.

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